

Thursday  
7 March 2024  
10-11 AM  
3-4 PM

## Lecture # 2

# PH102

## After Mid Semester Exams

by

Dr. Subhash Thota & Dr. Pankaj Kumar Mishra

Department of Physics, IIT Guwahati

Emails: [subhasht@iitg.ac.in](mailto:subhasht@iitg.ac.in)

[pankaj.mishra@iitg.ac.in](mailto:pankaj.mishra@iitg.ac.in)

# Chapter-5

## Magnetostatics

- The Lorentz Force Law, Biot-Savart Law
  - Ampere's Circuital Law

Ampere's Law (in Integral & differential form)

### Section 5.3 of Divergence and Curl of $\mathbf{B}$

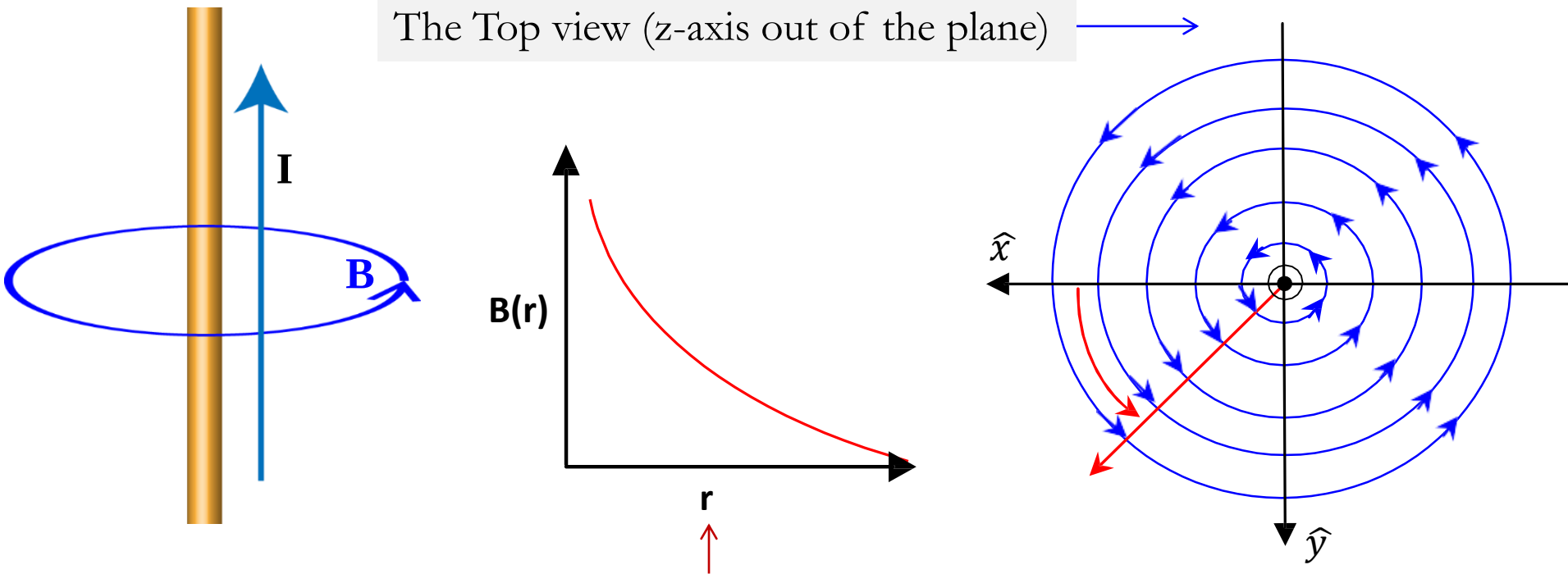
$$\nabla \cdot \vec{\mathbf{B}} = ???$$

$$\nabla \times \vec{\mathbf{B}} = ???$$

# The Divergence of $\mathbf{B}$

## Straight-Line Currents

Consider a straight line wire in which current ' $\mathbf{I}$ ' is coming out of page:



Variation of  $\mathbf{B}(r)$  with respect to radial distance for straight line wire

Clearly vector field ' $\mathbf{B}$ ' associated has non zero curl; Hence  $\nabla \times \vec{\mathbf{B}} \neq 0$

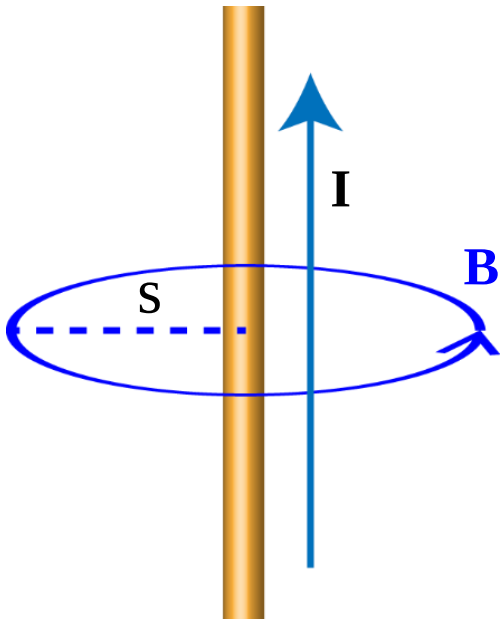
B-field has circulation (vorticity) associated with it !!!

Let's calculate the magnetic field associated with this straight line wire

The macroscopic magnetic field associated with this system is given by the relation:

$$\vec{B}(\vec{r}) = \left(\frac{\mu_0}{2\pi}\right) \frac{I}{s} \hat{\phi}$$

The integral of  $\vec{B}$  around circular path of radius 's'



$$\oint \vec{B} \cdot d\vec{l} = \oint \frac{\mu_0 I}{2\pi s} dl$$

$$= \frac{\mu_0 I}{2\pi s} \oint dl$$

$$= \frac{\mu_0 I}{2\pi s} 2\pi s$$

$$= \mu_0 I$$

Our answer is independent of  $s$

## Using Cylindrical co-ordinates

If we use the cylindrical coordinate  $(s, \phi, z)$  and current  $I$  is flowing along the  $z$  axis

$$dl = ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}$$

$$\vec{B}(\vec{r}) = \left(\frac{\mu_0}{2\pi}\right) \frac{I}{s} \hat{\phi}$$

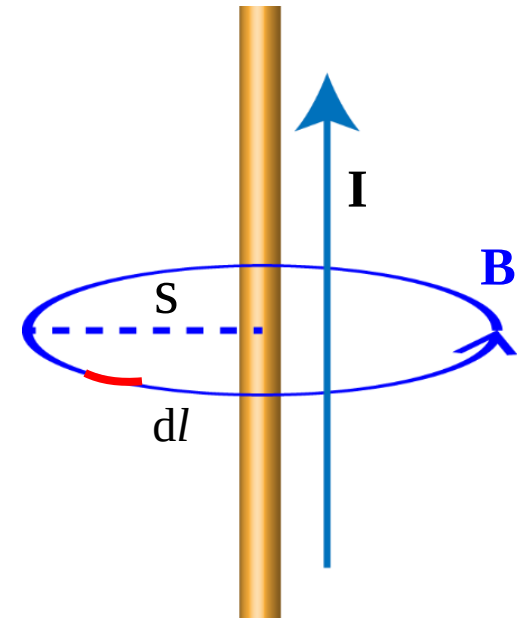
$$\vec{B} \cdot dl = \left(\frac{\mu_0}{2\pi}\right) \frac{I}{s} \hat{\phi} \cdot [ds \hat{s} + s d\phi \hat{\phi} + dz \hat{z}]$$

$$\vec{B} \cdot dl = \left(\frac{\mu_0}{2\pi}\right) \frac{I}{s} s d\phi$$

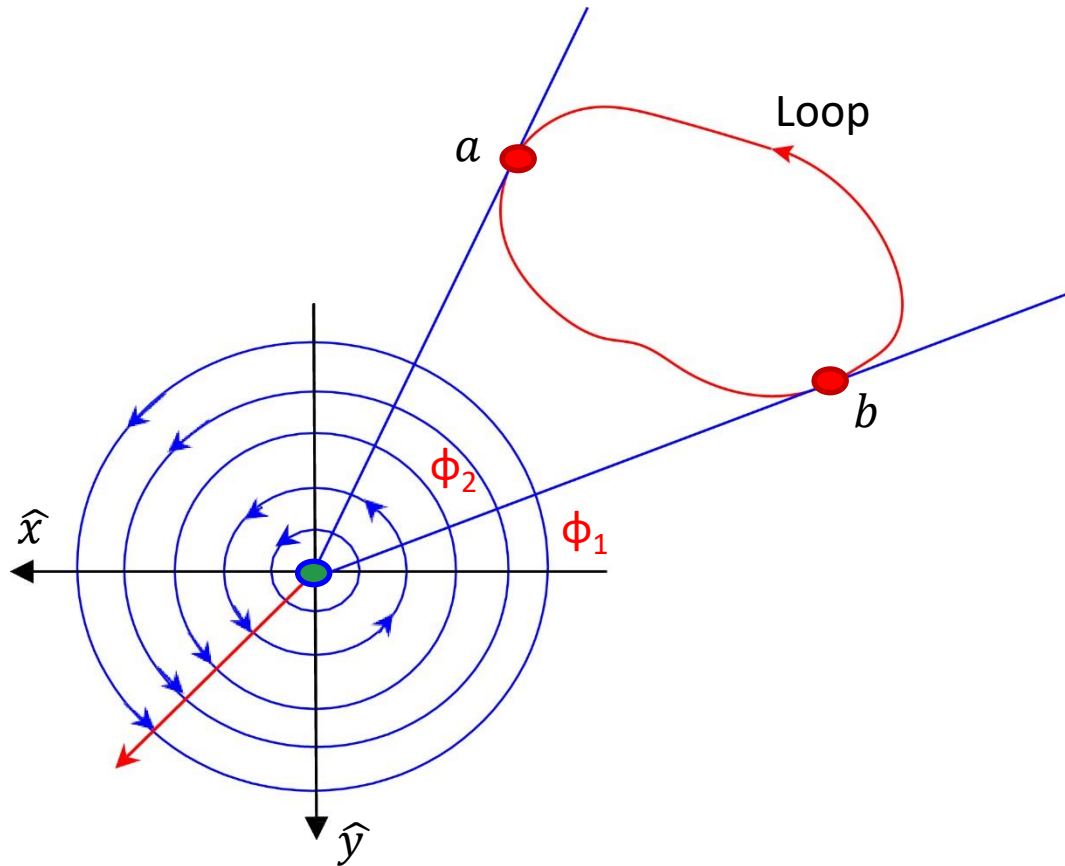
$$\oint \vec{B} \cdot dl = \frac{\mu_0 I}{2\pi} \oint \frac{1}{s} s d\phi = \frac{\mu_0 I}{2\pi} \int_0^{2\pi} \frac{1}{s} s d\phi$$

$$= \mu_0 I$$

$$\oint \vec{B} \cdot dl = \mu_0 I$$



If we had instead chosen a loop/contour/path which did not enclose (circle) the current, but rather, e.g. the contour shown in the figure below



$$\oint \vec{B} \cdot d\vec{l} = \int_a^b \hat{B}(r) \cdot d\vec{l}$$

$$= \frac{\mu_0 I}{2\pi} \int_{\phi_1}^{\phi_2} \frac{1}{r} d\phi$$

$$= 0$$

$$\oint \vec{B} \cdot d\vec{l} = 0$$

$\phi$  would go from  $\phi_1$  to  $\phi_2$  and back again so that the integration  $\int d\phi = 0$

Thus, we have obtained Ampere's Circuital Law (in integral form):

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enclosed}$$

$I_{enclosed}$  = Total current inclosed by the integration path

If the flow of charge is represented by a Volume current density  $\mathbf{J}$

$$\mathbf{J} = \frac{I_{enclosed}}{Area}$$

$$I_{enclosed} = \int \mathbf{J} \cdot d\mathbf{a}$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \oint_S \mathbf{J} \cdot d\mathbf{a}$$

## Applying Stokes Theorem

$$\oint \vec{B} \cdot d\vec{l} = \oint_S (\nabla \times \vec{B}) \cdot d\vec{a}$$

Substituting this value into the previous equation

$$\oint_S (\nabla \times \vec{B}) \cdot d\vec{a} = \mu_0 \oint_S \vec{J} \cdot d\vec{a}$$


$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \oint_S \vec{J} \cdot d\vec{a}$$

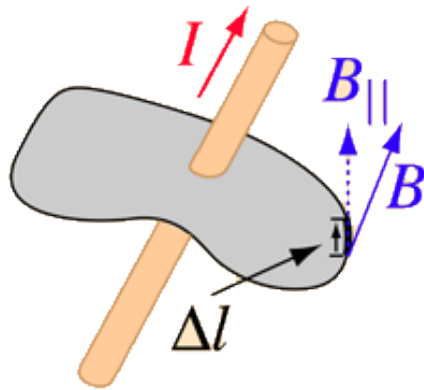
So the curl of  $\vec{B}$



$$\nabla \times \vec{B} = \mu_0 \vec{J}$$

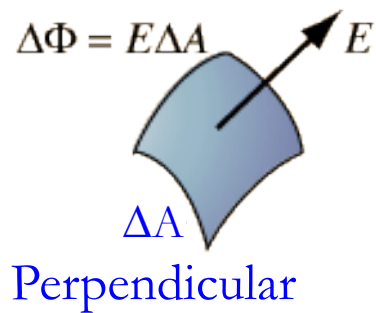
Thus, we have obtained Ampere's Circuital Law (in differential form)

Ampere's Law states that for any closed loop path, the sum of the length elements times the magnetic field in the direction of the length element is equal to the permeability times the electric current enclosed in the loop.

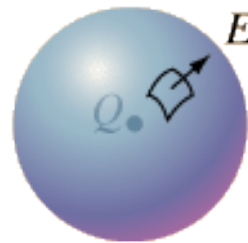


$$\sum B_{\parallel} \Delta l = \mu_0 I$$

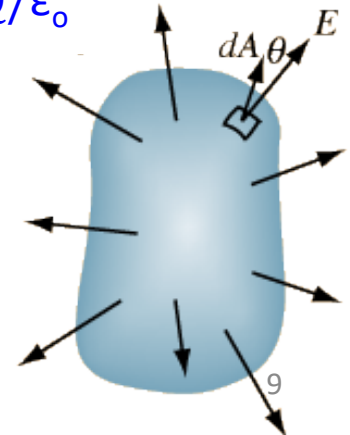
In the electric case, the relation of field to source is quantified in Gauss's Law which is a very powerful tool for calculating electric fields



$$\Phi_{\text{Electric}} = \frac{Q}{\epsilon_0}$$



$$\oint \mathbf{E} \cdot d\mathbf{A} = Q/\epsilon_0$$



# The Divergence of B

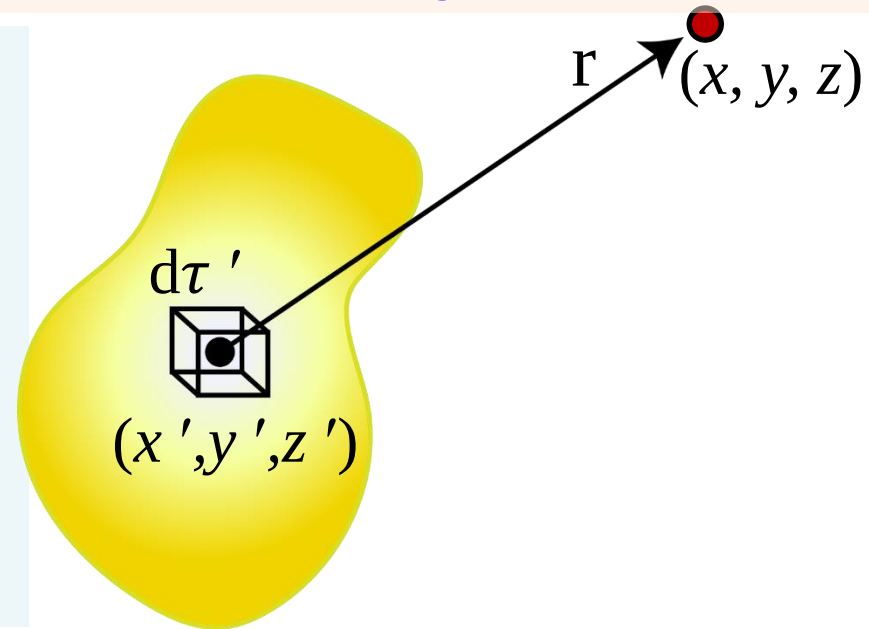
The Biot-Savart law for the general case of a volume current is given by

$$\vec{B}(r) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(r') \times \hat{r}}{r^2} d\tau' \quad \text{Eq. (1)}$$

This formula gives the B at a point  $r = (x, y, z)$  in terms of an integral over the current distribution  $J(x', y', z')$  as shown in the below figure.

## Important points to be remember:

- ❖ B is a function of  $(x, y, z)$ ,
- ❖ J is a function of  $(x', y', z')$
- ❖  $\vec{r} = (x-x')\hat{x} + (y-y')\hat{y} + (z-z')\hat{z}$
- ❖  $d\tau' = dx' dy' dz'$



The integration is taken over primed coordinates  $(x', y', z')$ , However the divergence and curl are to be taken with respect to the unprimed coordinates  $(x, y, z)$ .

Taking the divergence of the Biot-Savart equation i.e. Eq. 1

$$\nabla \cdot \vec{B}(r) = \frac{\mu_0}{4\pi} \int \nabla \cdot \left( \vec{J}(r') \times \frac{\hat{r}}{r^2} \right) d\tau' \quad \text{Eq. (2)}$$

Using the following relation

$$\nabla \cdot (\vec{A} \times \vec{B}) = \vec{B} \cdot (\nabla \times \vec{A}) - \vec{A} \cdot (\nabla \times \vec{B}) \quad \text{Eq. (3)}$$

Eq-2 can be rewritten as

$$\nabla \cdot \left( \vec{J}(r') \times \frac{\hat{r}}{r^2} \right) = \frac{\hat{r}}{r^2} \cdot (\nabla \times \vec{J}(r')) - \vec{J}(r') \cdot (\nabla \times \frac{\hat{r}}{r^2}) \quad \text{Eq. (4)}$$

Here  $\nabla \times \vec{J}(r') = 0$

Since  $\vec{J}$  doesn't depend on the unprimed variables (x, y, z)

Also we know that  $\nabla \times \frac{\hat{r}}{r^2} = 0$

Finally

$$\nabla \cdot \vec{B}(r) = 0$$

$$\nabla \cdot \vec{B}(r) = 0$$

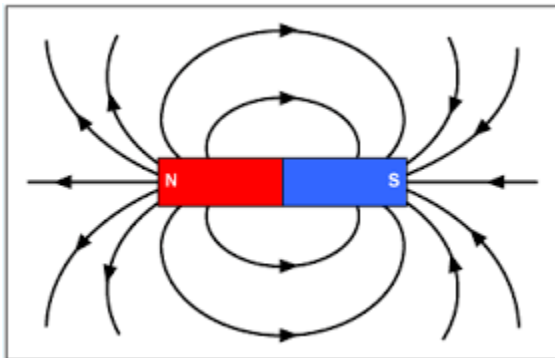
Evidently, the divergence of the magnetic field is zero.

### Significance

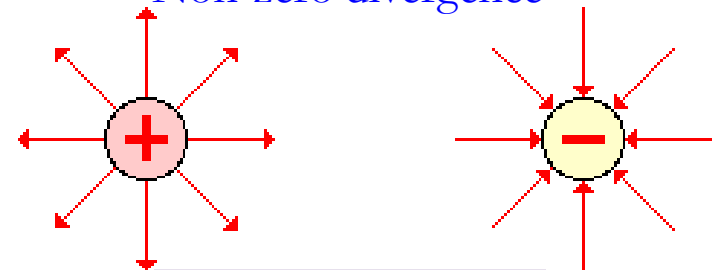
- ❑ The magnetic field is a solenoidal vector field.
- ❑ Also, it is equivalent to the statement that magnetic monopole (isolated North or South magnetic pole) does not exist.

The basic quantity for magnetism is the magnetic dipole, not the magnetic charge or monopole. Because no divergence of  $B$ .

Zero divergence  $\nabla \cdot \vec{B}(r) = 0$

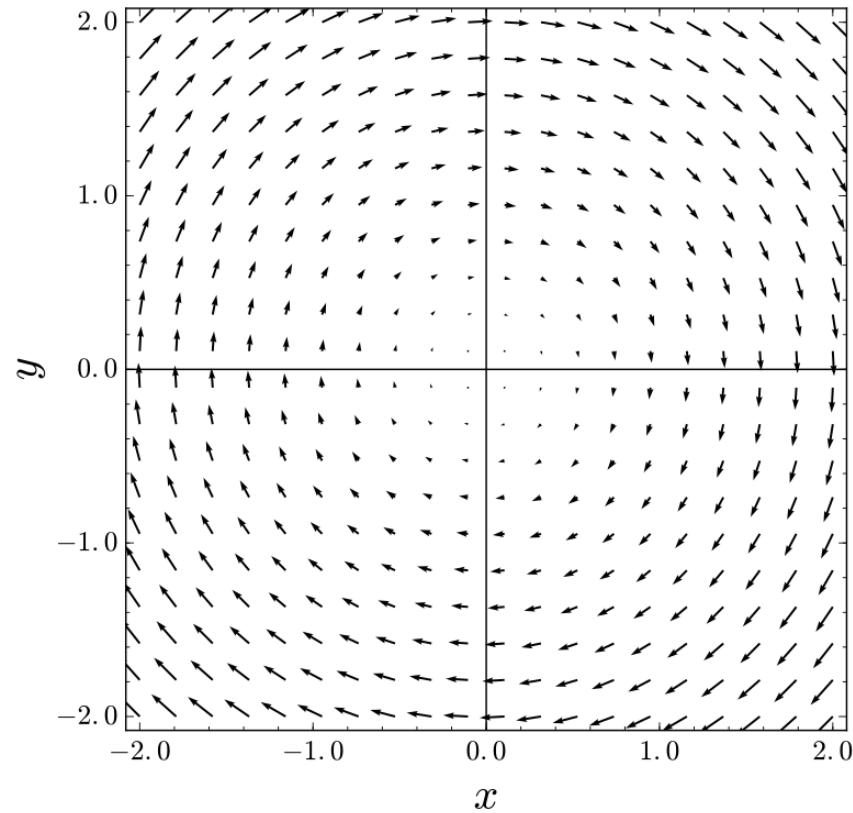


Non-zero divergence



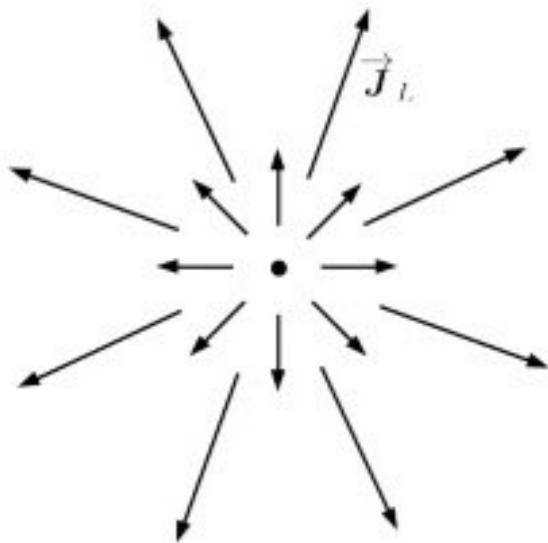
$$\nabla \cdot \vec{E}(r) \neq 0$$

## An example of a solenoidal vector field

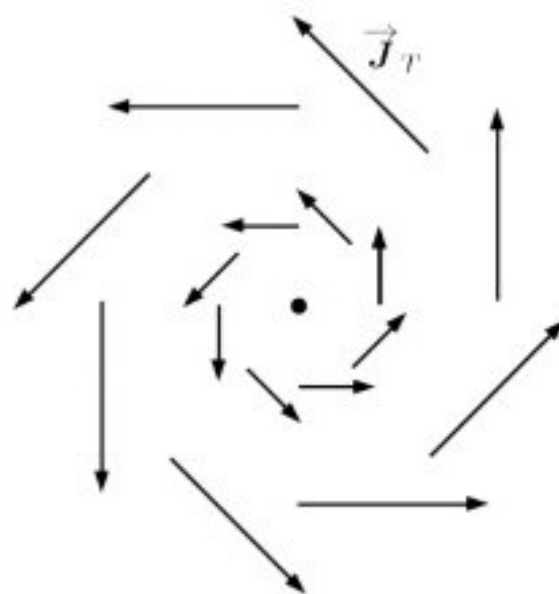


$$\nabla \cdot \vec{B}(r) = 0$$

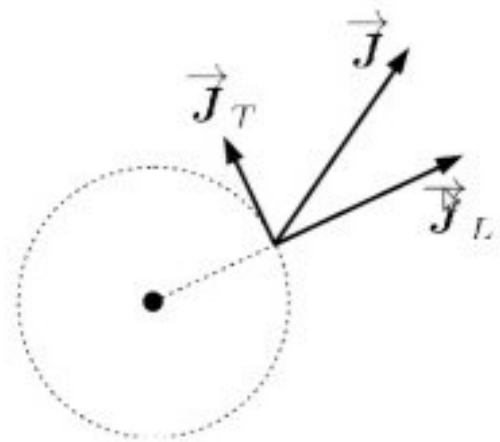
□ The magnetic field is a solenoidal vector field.



Irrotational or  
longitudinal current  
 $\vec{J}_L$



Solenoidal or tangential  
current  $\vec{J}_T$



Total current

# The Curl of B

The Biot-Savart law for the general case of a volume current is given by

$$\vec{B}(r) = \frac{\mu_0}{4\pi} \int \frac{\vec{J}(r') \times \hat{r}}{r^2} d\tau' \quad \text{Eq. (1)}$$

Taking the curl of the Biot-Savart equation i.e. Eq. 1

$$\nabla \times \vec{B}(r) = \frac{\mu_0}{4\pi} \int \nabla \times \left( \vec{J}(r') \times \frac{\hat{r}}{r^2} \right) d\tau' \quad \text{Eq. (5)}$$

Using the standard curl rule given by the following relation

$$\nabla \times (\vec{A} \times \vec{B}) = (\vec{B} \cdot \nabla) \vec{A} - (\vec{A} \cdot \nabla) \vec{B} + \vec{A}(\nabla \cdot \vec{B}) - \vec{B}(\nabla \cdot \vec{A}) \quad \text{Eq. (6)}$$

Consider

$$\nabla \times \left( \vec{J}(r') \times \frac{\hat{r}}{r^2} \right) \quad \vec{A} = \vec{J}(r') \quad \& \quad \vec{B} = \frac{\hat{r}}{r^2}$$

Substituting the values of A and B in the Eq. 6

$$\nabla \times (\vec{J}(r') \times \frac{\vec{\hat{r}}}{r^2}) = \left( \frac{\vec{\hat{r}}}{r^2} \cdot \nabla \right) \vec{J}(r') - (\vec{J}(r') \cdot \nabla) \frac{\vec{\hat{r}}}{r^2} + \vec{J}(r') \left( \nabla \cdot \frac{\vec{\hat{r}}}{r^2} \right) - \frac{\vec{\hat{r}}}{r^2} (\nabla \cdot \vec{J}(r'))$$

We know that

$$\nabla \cdot \vec{J}(r') = 0$$

$$\left( \frac{\vec{\hat{r}}}{r^2} \cdot \nabla \right) \vec{J}(r') = 0$$

Since  $\mathbf{J}$  doesn't depend on the unprimed variables (x, y, z)

Now, the equation becomes:

$$\nabla \times (\vec{J}(r') \times \frac{\vec{\hat{r}}}{r^2}) = \vec{J}(r') \left( \nabla \cdot \frac{\vec{\hat{r}}}{r^2} \right) - (\vec{J}(r') \cdot \nabla) \frac{\vec{\hat{r}}}{r^2}$$

Eq. (7)

## Integration of the second term of equation 7 become zero

Proof:

We can switch from  $\nabla$  to  $\nabla'$  at the cost of a minus sign:

Since  $\mathbf{r}$  depends only on the difference between the coordinates

$$-(\vec{J} \cdot \nabla) \frac{\hat{r}}{r^2} = (\vec{J} \cdot \nabla') \frac{\hat{r}}{r^2}$$

$$\because \frac{\partial}{\partial x} f(\mathbf{x}-\mathbf{x}') = - \frac{\partial}{\partial x'} f(\mathbf{x}-\mathbf{x}')$$

Rewriting the eq-7

$$(\vec{J} \cdot \nabla') \left( \frac{\hat{r}}{r^2} \right) = \nabla' \cdot \left[ \frac{\hat{r}}{r^2} \vec{J} \right] - \left( \frac{\hat{r}}{r^2} \right) (\nabla' \cdot \vec{J})$$

$$\because \nabla \cdot (f \vec{A}) = f (\nabla \cdot \vec{A}) + \vec{A} \cdot (\nabla f)$$

= 0 for steady state currents

$$(\vec{J} \cdot \nabla') \left( \frac{\hat{r}}{r^2} \right) = \nabla' \cdot \left[ \frac{\hat{r}}{r^2} \vec{J} \right]$$

$$= \frac{\partial \rho_{free}}{\partial t} = 0$$

$$\oint_V \nabla' \cdot \left[ \frac{\hat{r}}{r^2} J \right] d\tau' \quad \longrightarrow$$

$$\oint_S \left[ \frac{\hat{r}}{r^2} \right] J \cdot da' \quad \longleftarrow$$

Using Divergence theorem

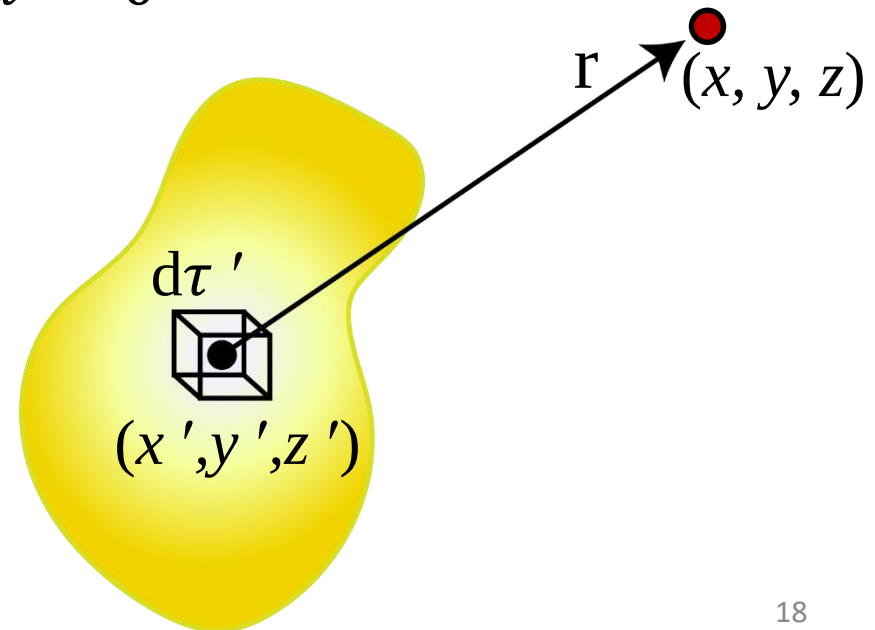
On the boundary the current is zero (all current is safely inside) and hence the surface integral is zero

$$\oint_S \left[ \frac{\hat{r}}{r^2} \right] J \cdot da' = 0$$

At the surface

$$J = 0$$

Because all current is safely enclosed inside the loop

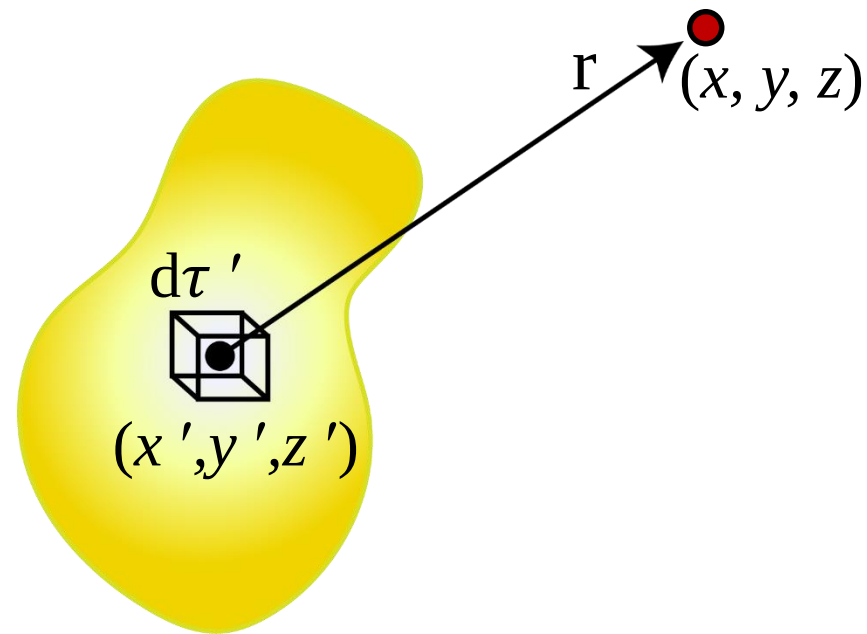


It's the volume that appears in the Biot-Savart law - large enough, that is, to include all the current. You can make it bigger than that, if you like;  $J = 0$  out there anyway, so it will add nothing to the integral. The essential point is that on the boundary the current is zero (all current is safely inside) and hence the surface integral (5.53) vanishes.

At the surface

$$J = 0$$

Because all current is safely  
enclosed inside the loop



If  $J$  itself extends to infinity (as in the case of an infinite straight wire), the surface integral is still zero, though the analysis calls for greater care.

$$\nabla \times \left( \vec{J}(r') \times \frac{\vec{\hat{r}}}{r^2} \right) = \vec{J}(r') \left( \nabla \cdot \frac{\vec{\hat{r}}}{r^2} \right) - \underbrace{\left( \vec{J}(r') \cdot \nabla \right) \frac{\vec{\hat{r}}}{r^2}}_{=0} \quad \text{Eq. 7}$$

Substituting in Eq. 5

$$\nabla \times \vec{B}(r) = \frac{\mu_0}{4\pi} \int \nabla \times \left( \vec{J}(r') \times \frac{\vec{\hat{r}}}{r^2} \right) d\tau' \quad \text{Eq. 5}$$

$$\nabla \times \vec{B}(r) = \frac{\mu_0}{4\pi} \int \left( \vec{J}(r') 4\pi \delta^3(r - r') \right) d\tau'$$

Using the relation

$$\nabla \cdot \frac{\vec{\hat{r}}}{r^2} = 4\pi \delta^3(r)$$

$$\nabla \times B = \mu_0 J(r)$$

Ampere's Law (in differential form)

This confirms that this equation is not restricted to straight-line currents, but holds quite generally in magneto-statics.

# Applications of Ampere's Law

□ Ampere's law in Differential form :

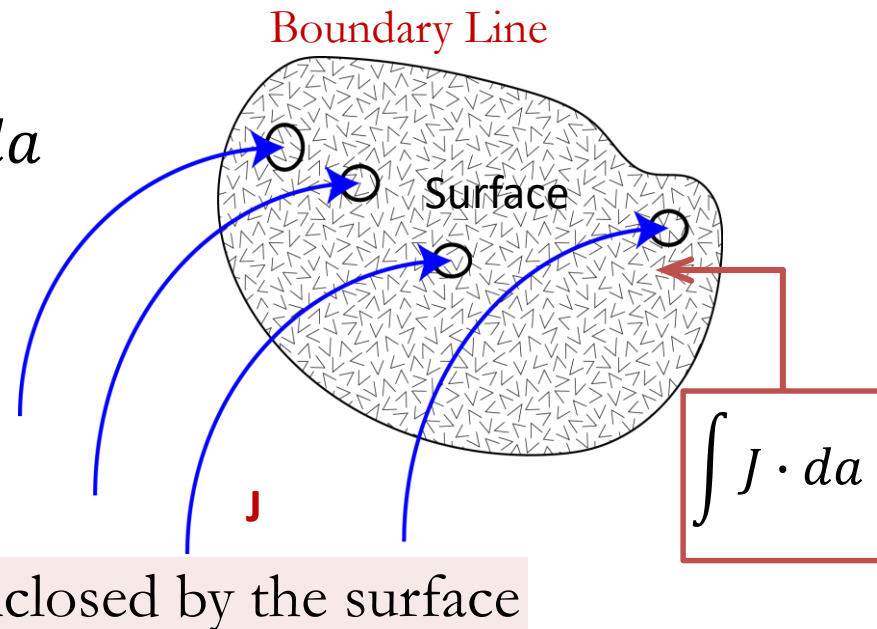
$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J}(r)$$

□ Ampere's law in Integral form :

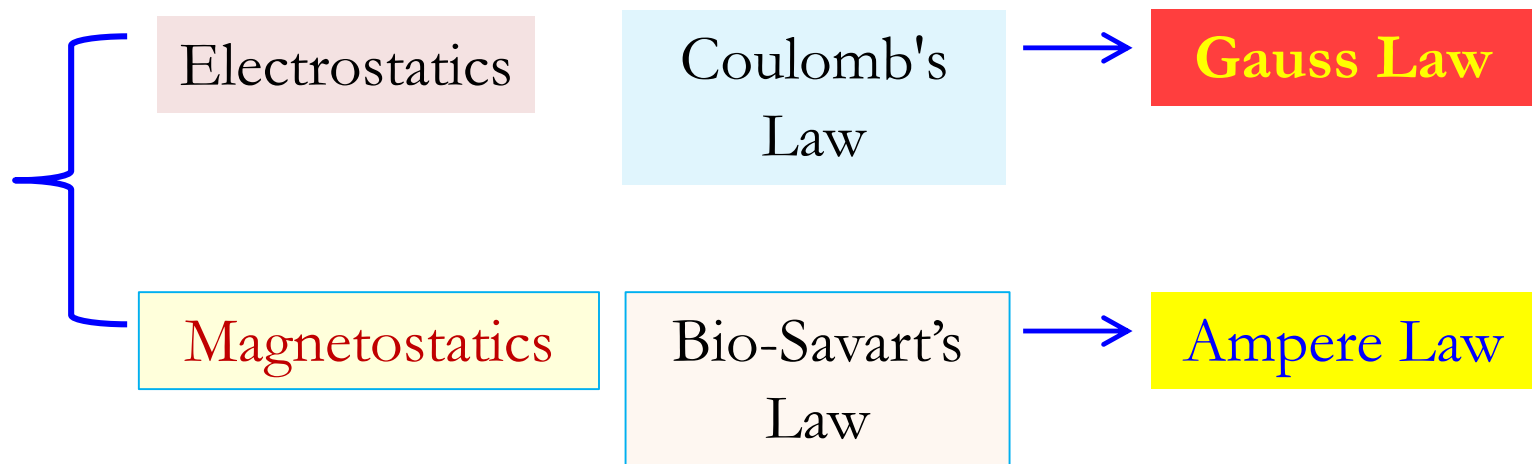
Applying Stokes theorem to the Differential form of Ampere's law.

$$\int \nabla \times \mathbf{B} \cdot d\mathbf{a} = \oint \vec{B} \cdot d\mathbf{l} = \mu_0 \int \mathbf{J} \cdot d\mathbf{a}$$

$$\oint \vec{B} \cdot d\mathbf{l} = \mu_0 I_{\text{enclosed}}$$



Just as the Biot-Savart law plays a role in magneto-statics that Coulomb's law assumed in electrostatics, so Ampere's plays the role of Gauss's:



In particular, for currents with appropriate symmetry, Ampere's law in integral form offers a lovely and extraordinarily efficient means for calculating the magnetic field.

## Maxwell's Equations for Electrostatics ( Differential Form )



$$\vec{\nabla} \cdot \vec{E}(\vec{r}) = \frac{\rho_{TOT}(\vec{r})}{\epsilon_0}$$

(Gauss' Law) with

$$\rho_{TOT}(\vec{r}) = \rho_{free}(\vec{r}) + \rho_{bound}(\vec{r})$$

$$\vec{\nabla} \times \vec{E}(\vec{r}) = 0 \quad (always) \leftarrow \vec{E}(\vec{r}) \text{ associated with conservative electrostatic force}$$

## Maxwell's Equations for Magnetostatics ( Differential Form )



$$\vec{\nabla} \cdot \vec{B}(\vec{r}) = 0$$

$\leftarrow$  no magnetic charges/ no magnetic monopoles

$$\vec{\nabla} \times \vec{B}(\vec{r}) = \mu_0 \vec{J}(\vec{r})$$

(Ampere's Law)

Force Law for Electric Charges

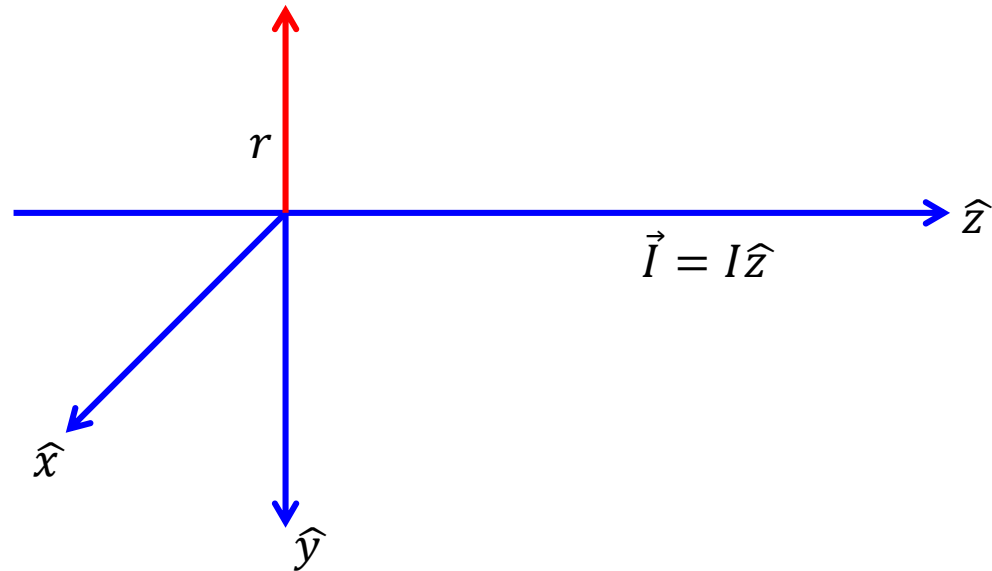
$$\vec{F}_e(\vec{r}) = Q(\underbrace{\vec{E}(\vec{r})}_{\text{Coulomb's Law}} + \underbrace{\vec{v}(\vec{r}) \times \vec{B}(\vec{r})}_{\text{Lorentz Force Law}})$$

Coulomb's Law    Lorentz Force Law

## Examples

**Example-1:** Determine the macroscopic magnetic field  $\vec{B}(r)$  at a perpendicular distance  $r$  away from infinitely long, straight wire carrying steady current  $\mathbf{I}$ .

Using the integral form of  
Ampere's law



**Solution-1:** We know that  $\hat{B} \parallel \hat{\phi}$

Use the integral form of Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$dl = r d\phi$$

Using arc-length relation

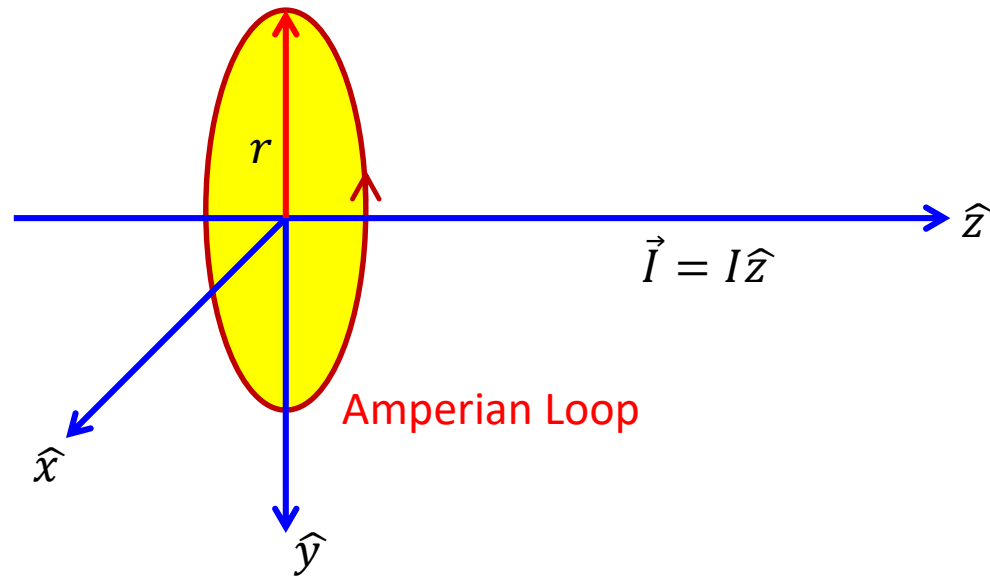
$$\hat{B} \parallel \hat{\phi} \quad \hat{\phi} \parallel d\vec{l}$$

$$B(r) \parallel d\vec{l}$$

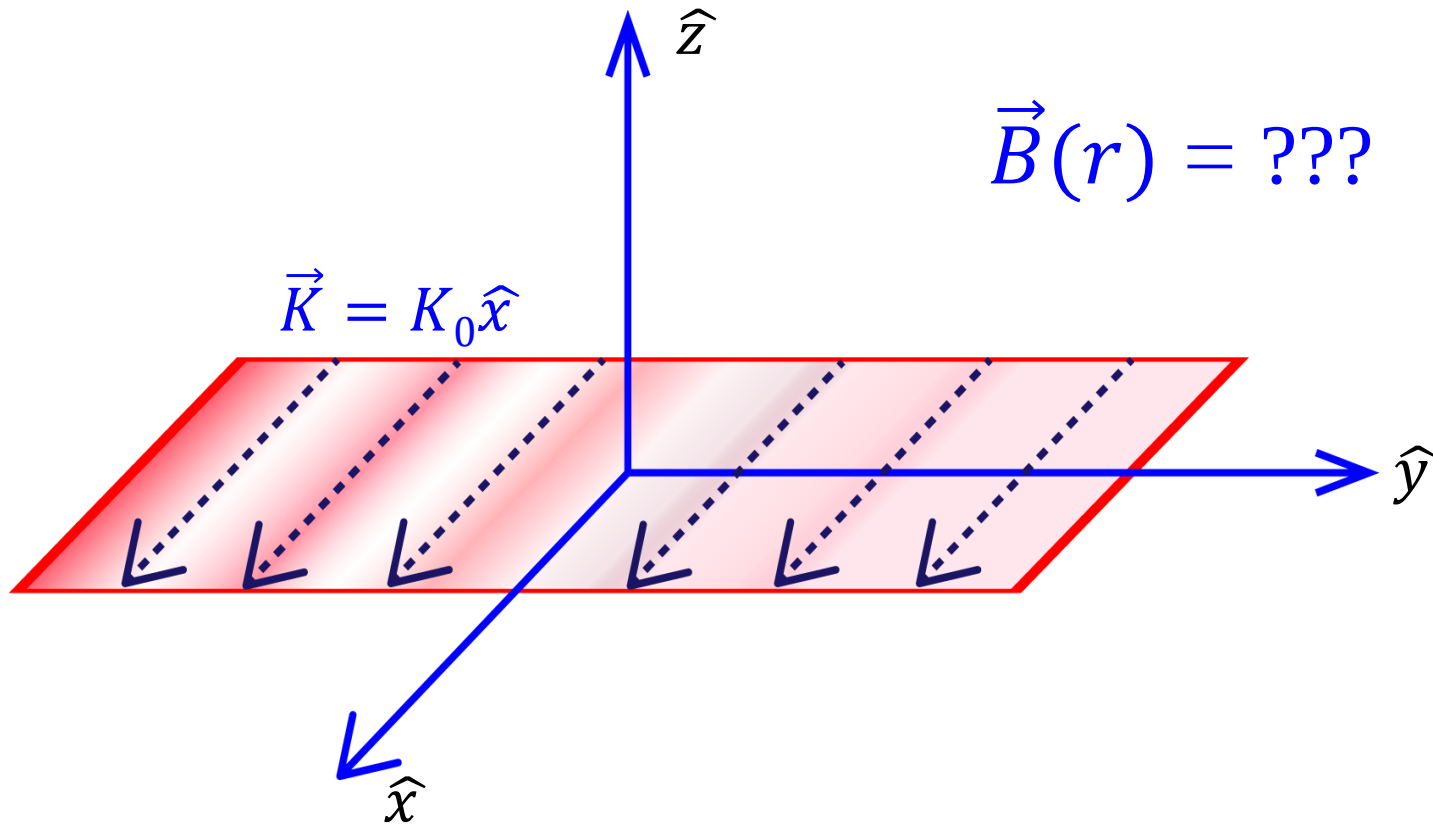
$$\vec{B}(r) \cdot d\vec{l} = \vec{B}(r) \cdot r d\phi$$

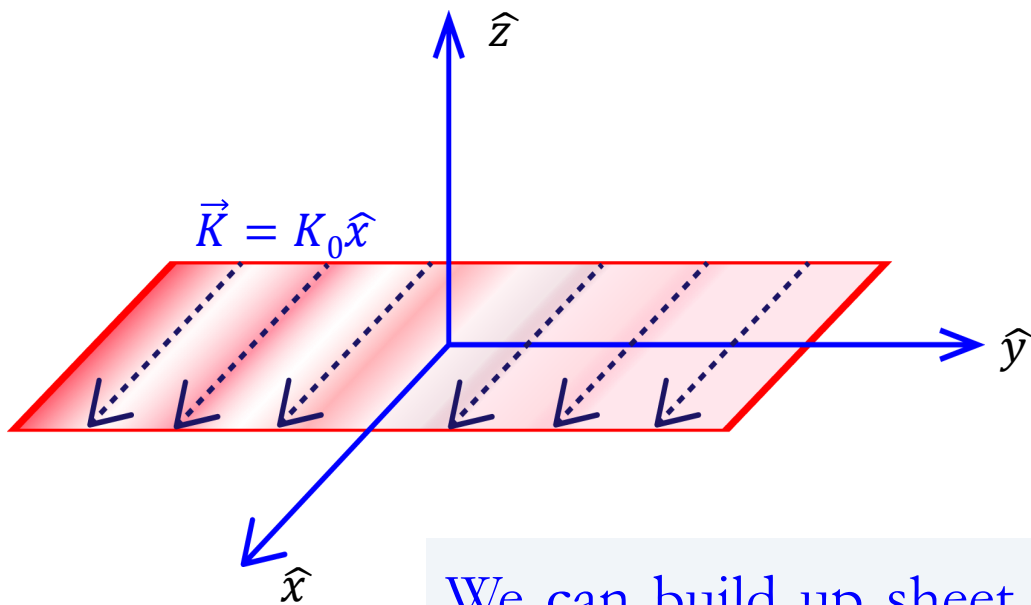
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}} = \int_{\phi=0}^{\phi=2\pi} \vec{B}(r) \cdot r d\phi = 2\pi r B(r) = \mu_0 I$$

$$\vec{B}(\vec{r}) = \left(\frac{\mu_0}{2\pi}\right) \frac{I}{r} \hat{\phi}$$

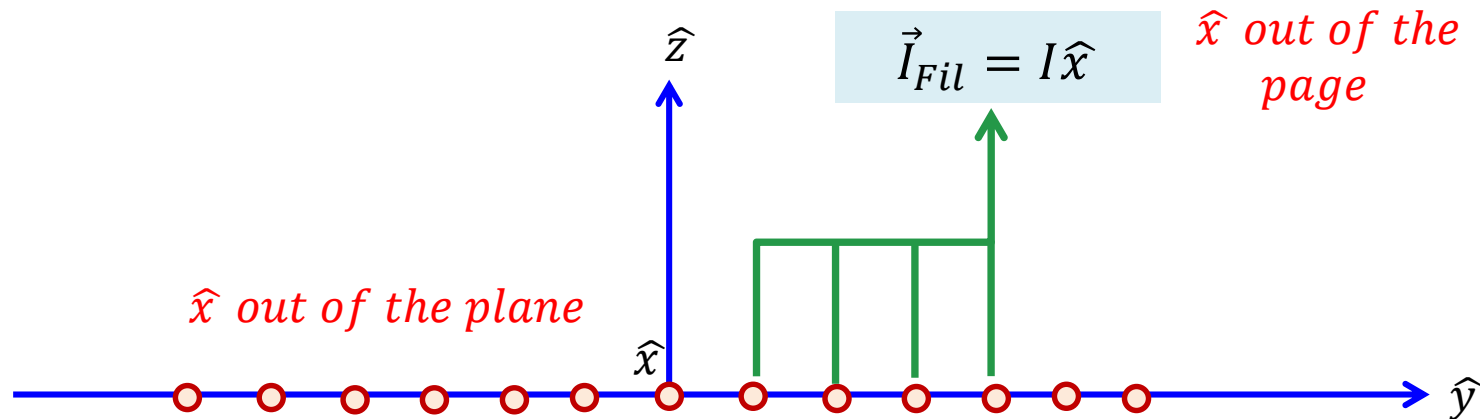


**Example-2:** Find the magnetic field  $\vec{B}(\mathbf{r})$  associated with an infinite planar sheet of uniform surface current,  $\vec{K} = K_0 \hat{x}$  Amperes/meter flowing over the x-y plane.





We can build up sheet current  $\vec{K} = K_0 \hat{x}$  from a linear superposition of infinitesimal filamentary line currents:



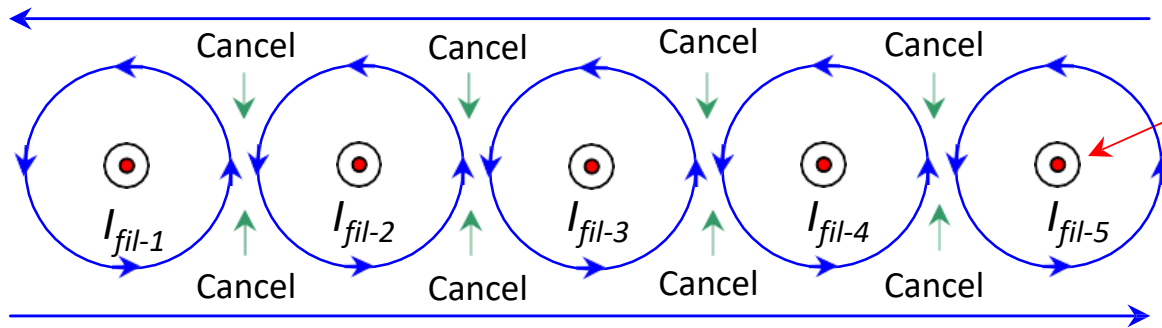
End-View of Uniform Surface/ “Sheet” Current

# Linear superposition of infinitesimal filamentary line currents

## Microscopic View of Filamentary/Infinitesimal Line Currents

$$\vec{I}_{Fil} = I \hat{x}$$

Out of the page



$$\vec{B}^{net}_{above}(z > 0) = -B_0 \hat{y}$$

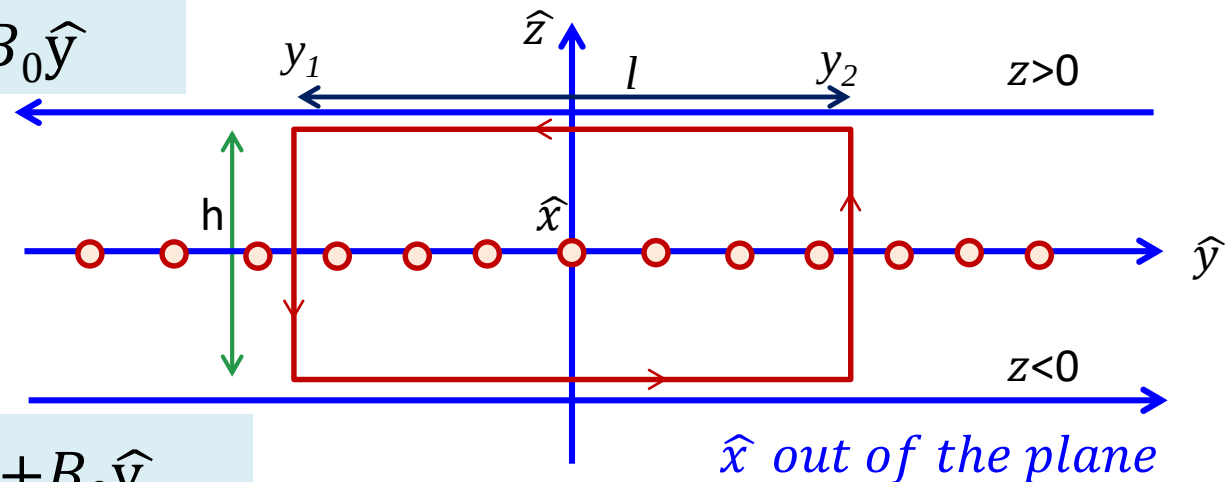
$$\vec{B}^{net}_{below}(z < 0) = +B_0 \hat{y}$$

Take a contour as shown below

$$\vec{B}_{Net-above} = -B_0 \hat{y}$$

$$\vec{K} = K_0 \hat{x} \text{ A/m}$$

$$\vec{B}_{Net-below} = +B_0 \hat{y}$$

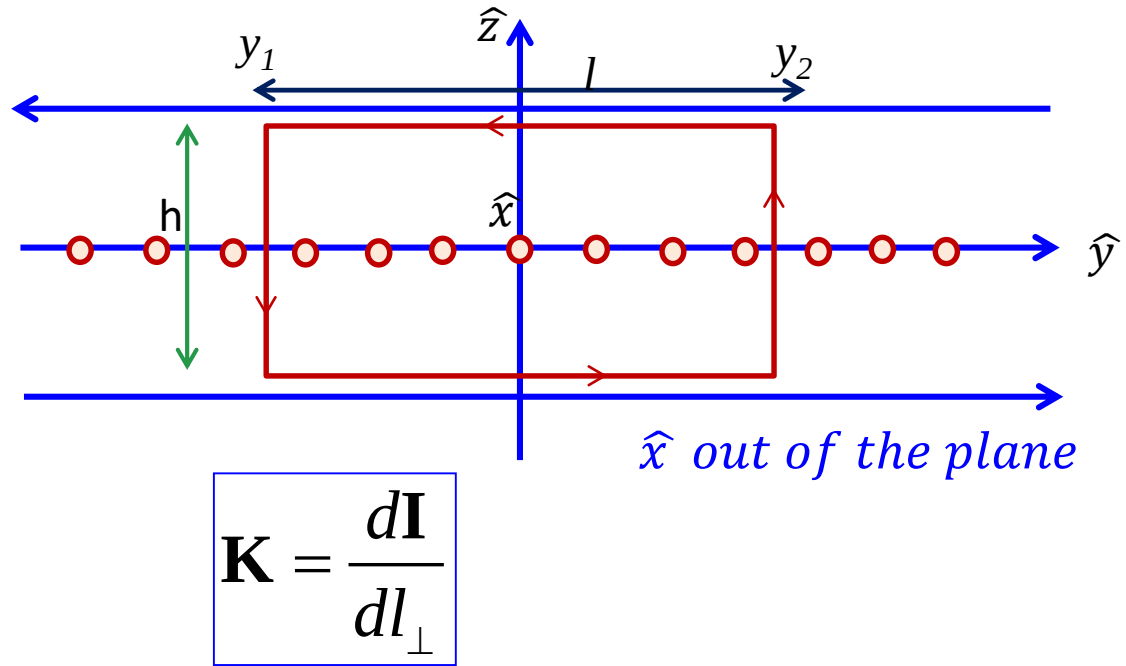


$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$I_{\text{enclosed}} = \int_{l_{\perp}=y_1}^{l_{\perp}=y_2} \vec{K} d l_{\perp}$$

$$= \int_{y_1}^{y_2} K_0 \hat{x} d l_{\perp}$$

$$= K_0 \hat{x} l$$



As  $h \rightarrow 0$ , only horizontal portions of the contour loop will remain

$$\begin{aligned} \oint \vec{B}(r) \cdot d\vec{l} &= \mu_0 I_{\text{enclosed}} \\ &= \mu_0 K_0 l \end{aligned}$$

L.H.S

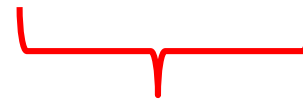
$$\oint \vec{B}(r) \cdot d\vec{l} = \int_{y_2}^{y_1} \vec{B}_{above} \cdot d\vec{l} + \int_{y_1}^{y_2} \vec{B}_{below} \cdot d\vec{l}$$

$$= \int_{y_2}^{y_1} -B_0 \hat{y} \, dl + \int_{y_1}^{y_2} B_0 \hat{y} \, dl$$

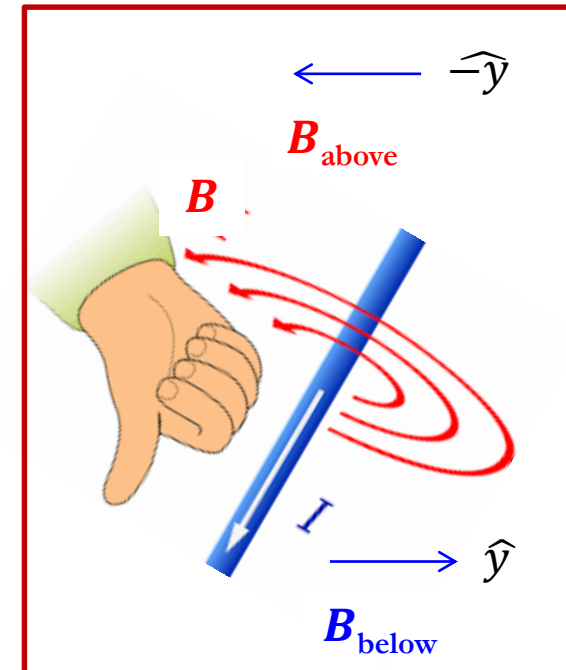
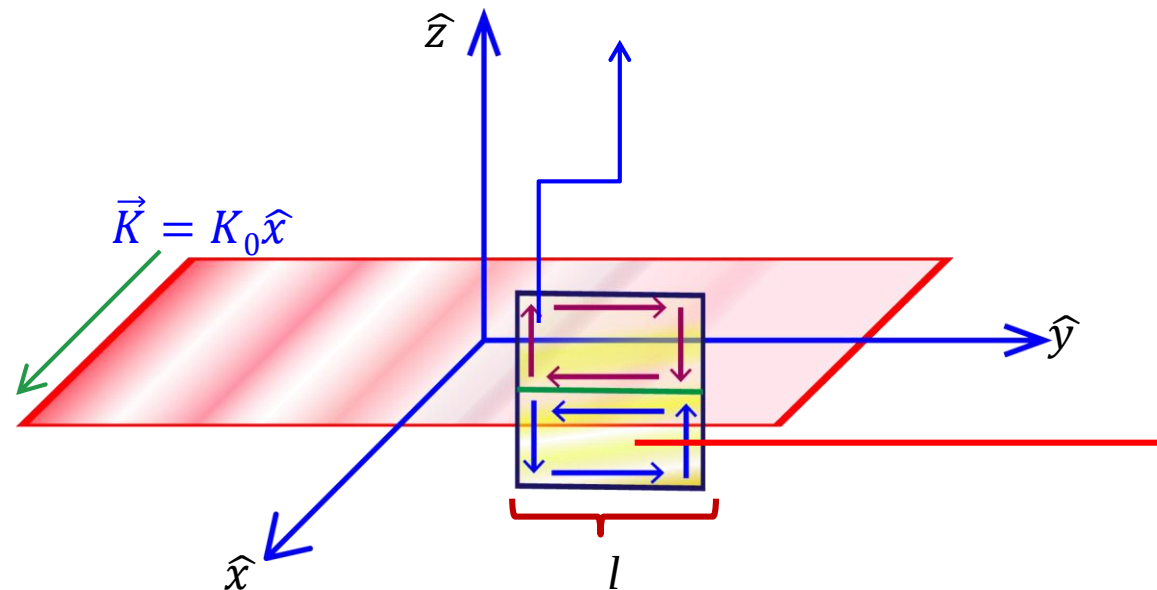
$$= (-B_0 \hat{y}) \cdot (-l \hat{y}) + (B_0 \hat{y}) \cdot (l \hat{y})$$



Above



Below

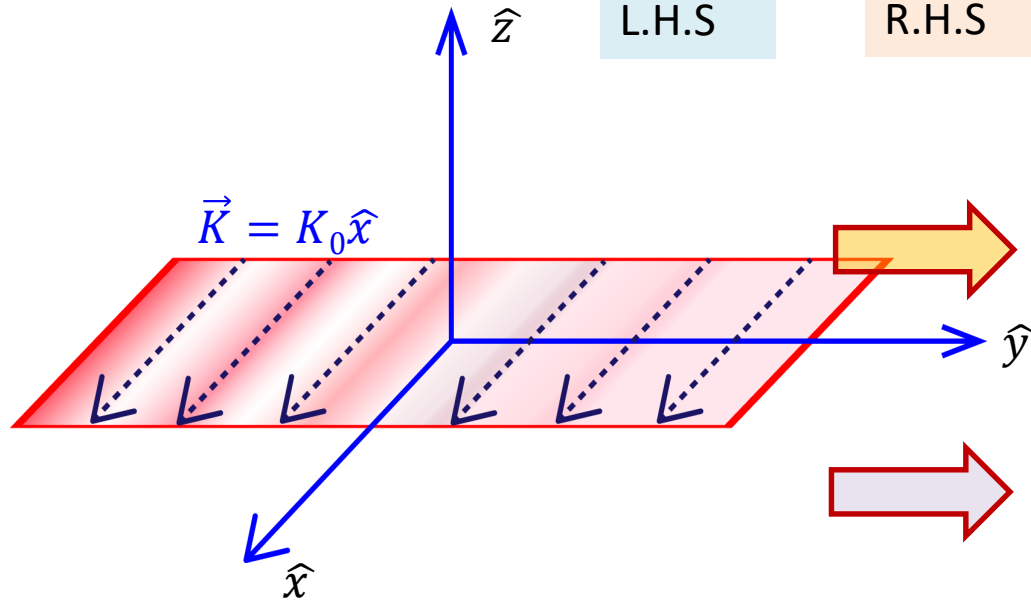


$$= (B_0 l) + (B_0 l)$$

$$= 2B_0 l = \mu_0 K_0 l$$

L.H.S

R.H.S



$$\vec{B}_{above} = -\frac{1}{2} \mu_0 K_0 \hat{y}$$

$$\vec{B}_{below} = +\frac{1}{2} \mu_0 K_0 \hat{y}$$

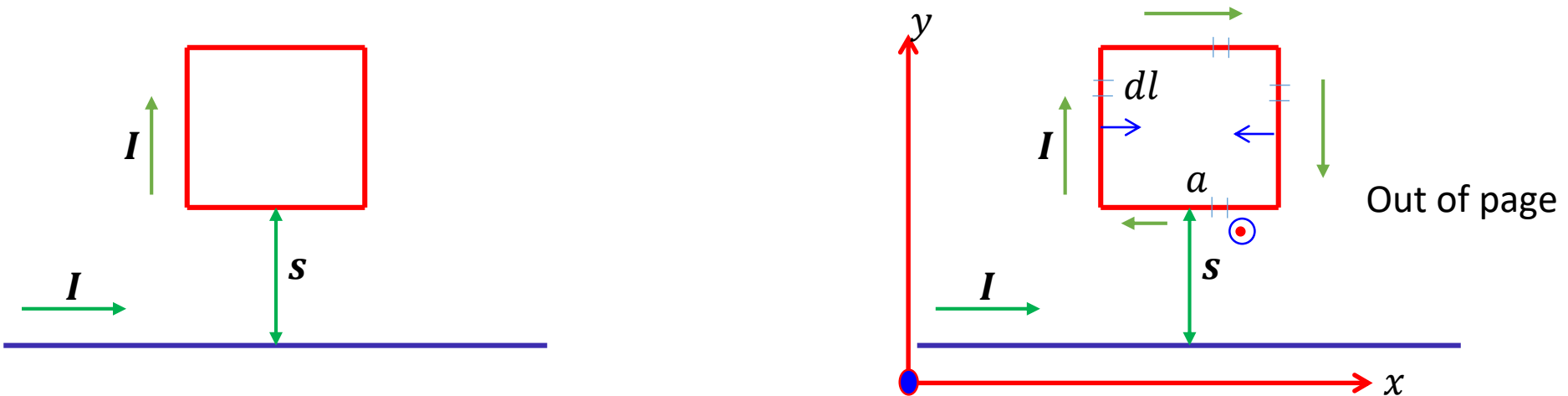
## Analogy to Electrostatics

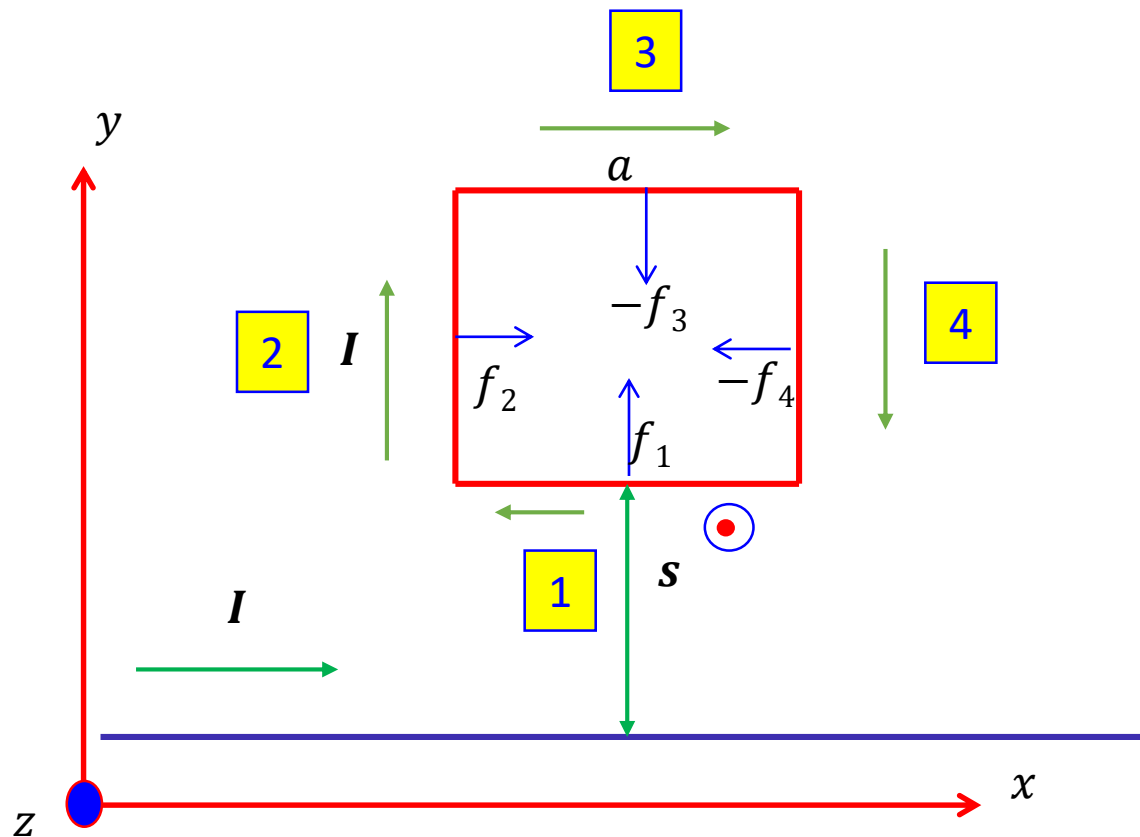
Completely analogous to what we found for the field associated with an infinite planar electrically charged conducting sheet of uniform surface charge density  $\sigma_0$  Coulombs/m<sup>2</sup>

Note that for the infinite planar sheet current, that the  **$\vec{B}$ -field is independent of the height  $z$**  of the observer above (or below) the sheet current

### Example-3

Find the force on a square loop placed as shown in Figure near an infinite straight wire. The loop and the wire carry a steady current  $I$ .





$$F = \cancel{f_2} - \cancel{f_4} + f_1 - f_3$$

$$F = f_1 - f_3$$

Magnetic field due to straight current carrying wire (segment-1), at a distance  $s$  is :  $B = \frac{\mu_0 I}{2\pi s}$

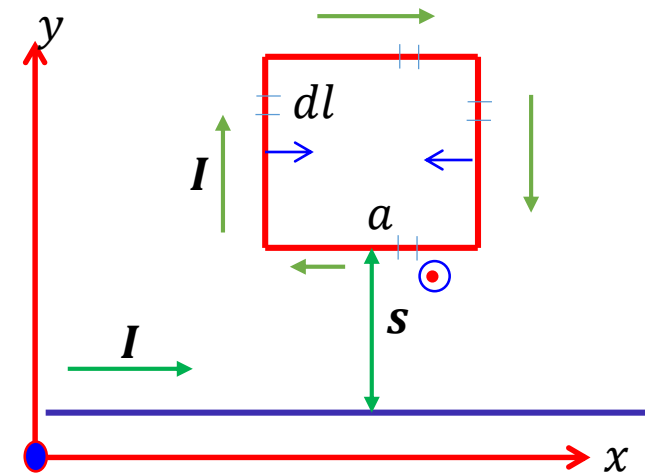
Field On segment-3  $\rightarrow B = \frac{\mu_0 I}{2\pi(s+a)}$

Force On segment-1  $\mathbf{F} = \int I (\vec{dl} \times \vec{B})$

$$f_1 = \int I (\vec{dl} \times \frac{\mu_0 I}{2\pi s} \hat{z})$$

$$= \int_0^a I (-dx \hat{x} \times \frac{\mu_0 I}{2\pi s} \hat{z})$$

$$f_1 = \frac{\mu_0 a I^2}{2\pi s} \hat{y}$$



$$\hat{x} \times \hat{z} = -\hat{y}$$

Force On segment-3  $F = \int I (\vec{dl} \times \vec{B})$

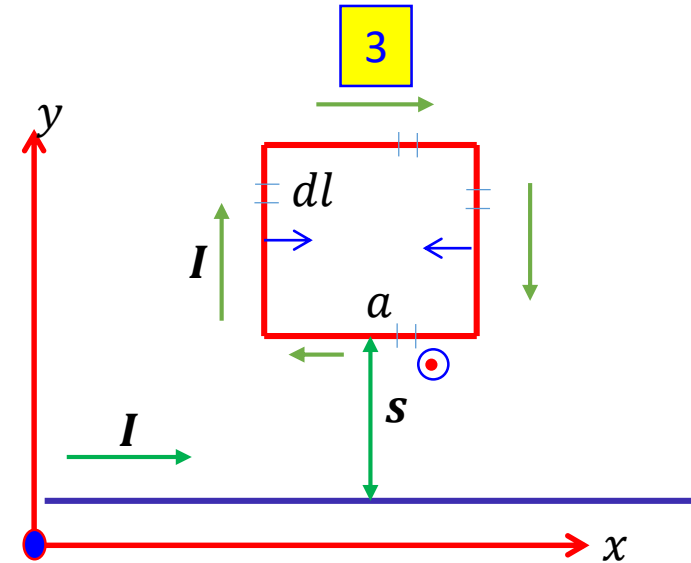
$$f_3 = \int I (\vec{dx} \hat{x} \times \frac{\mu_0 I}{2\pi(s+a)} \hat{z})$$

$$f_3 = \frac{\mu_0 I^2}{2\pi(s+a)} \int (\vec{dx} \hat{x} \times \hat{z})$$

$$f_3 = -\frac{\mu_0 I^2}{2\pi(s+a)} \int_0^a (\vec{dx} \hat{y})$$

$$f_3 = -\frac{\mu_0 a I^2}{2\pi(s+a)} \hat{y}$$

$$\hat{x} \times \hat{z} = -\hat{y}$$



These forces are directed opposite each other just as the right-hand rule indicated. The total force on the square loop is,

$$F = f_1 - f_3$$

$$F = \frac{\mu_0 a I^2}{2\pi s} \hat{y} - \frac{\mu_0 a I^2}{2\pi(s+a)} \hat{y}$$

$$F = \frac{\mu_0 a I^2}{2\pi} \left( \frac{1}{s} - \frac{1}{s+a} \right) \hat{y}$$

Since,  $1/s > 1/(s+a)$  it is seen that the net force is away from the infinite wire.

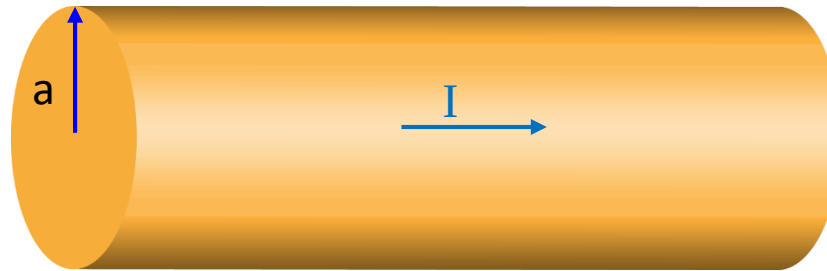
### Example-4

A steady current  $I$  flows down a long cylindrical wire of radius  $a$ . Find the magnetic field, both inside and outside the wire, if

(a) The current is uniformly distributed over the outside surface of the wire.

(b) The current is distributed in such a way that  $J$  is proportional to  $s$ , the distance from the axis.

Sol :



(a) Apply Ampere's circuital law ,  $\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{enc}$

$$\text{For } s < a \quad I_{enc} = 0 \quad \rightarrow \quad \vec{B} = 0$$

( here  $s$  is any distance from the axis of the cylinder )

$$\text{For } s > a \quad I_{enc} = I \quad \rightarrow \quad \vec{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}$$

(b) Given ,  $J = ks$  where  $k =$  proportionality constant

$$I = \int_0^a J da = \int_0^a (ks)(2\pi s ds) = 2\pi k \frac{a^3}{3}$$

$$\text{or, } k = \frac{3I}{2\pi a^3}$$

For  $s > a$   $I_{enc} = I \rightarrow \boxed{\vec{B} = \frac{\mu_0 I}{2\pi s} \hat{\phi}}$

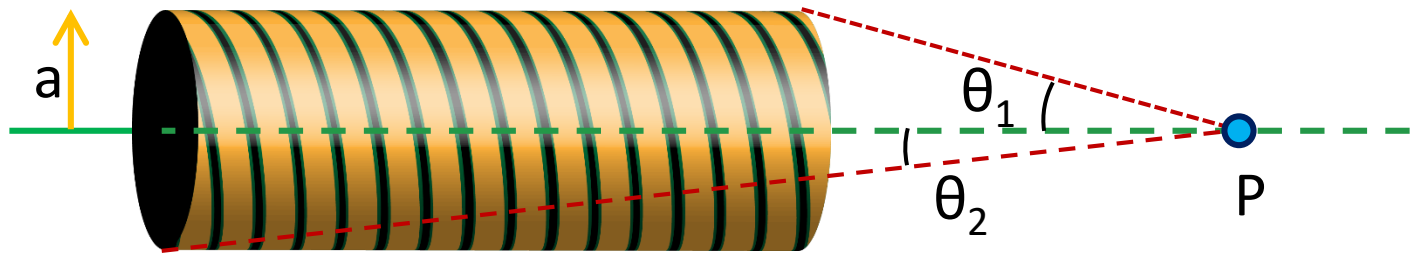
For  $s < a$   $I_{enc} = \int_0^s J da = \int_0^s (ks')(2\pi s' ds')$

$$= 2\pi k \frac{s^3}{3} = I \frac{s^3}{a^3} \quad (\text{Putting value of } k)$$

$$\boxed{\vec{B} = \frac{\mu_0 I}{2\pi} \frac{s^2}{a^3} \hat{\phi}}$$

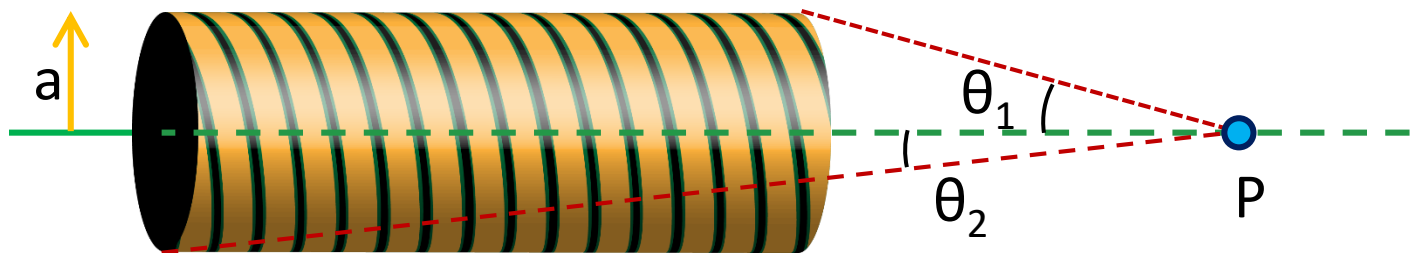
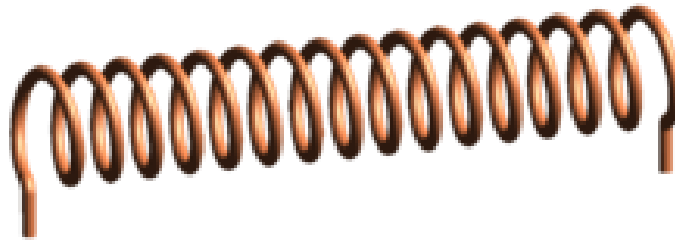
Example-1: Find the magnetic field at point P on the axis of a tightly wound solenoid (helical coil) consisting of 'n' turns per unit length wrapped around a cylindrical tube of radius 'a' and carrying current 'I'.

Express your answer in terms of  $\theta_1$  and  $\theta_2$  (it's easiest that way). Consider the turns to be essentially circular. What is the field on the axis of an infinite solenoid (infinite in both directions)?

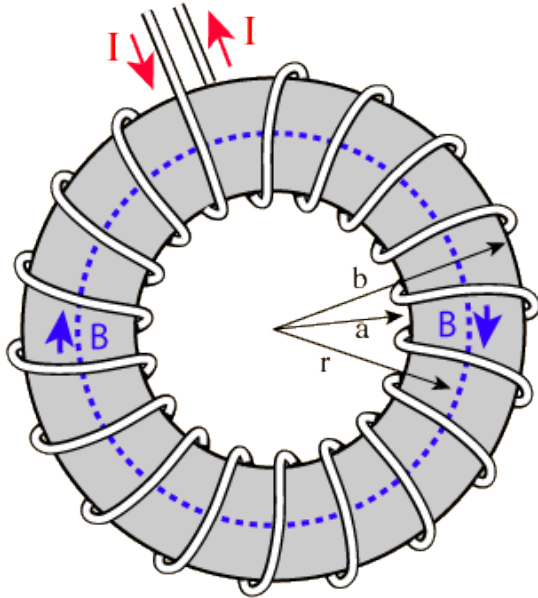


Before solving the problem let's see the general introduction to the Solenoid and Toroid

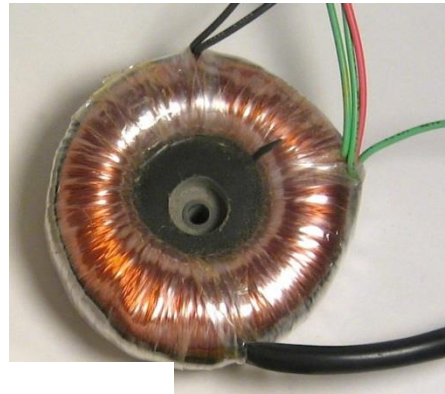
A solenoid (from the French solénoïde, derived in turn from the Greek solen "Pipe, channel" + combining form of Greek eidos "form, shape"



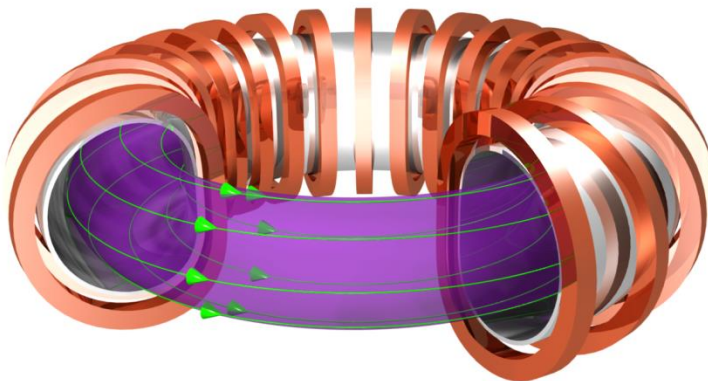
# Toroid



A **toroid** is a coil of insulated or enameled wire wound on a donut-shaped form made of powdered iron. A **toroid** is used as an inductor in electronic circuits, especially at low frequencies where comparatively large inductances are necessary.



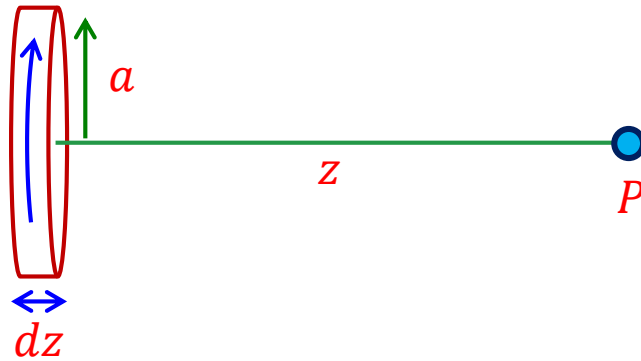
A small toroidal transformer.



A **tokamak** is a device that uses magnetic field to confine plasma in the shape of a torus

The tokamak is one of several types of magnetic confinement devices

produces controlled thermonuclear fusion power



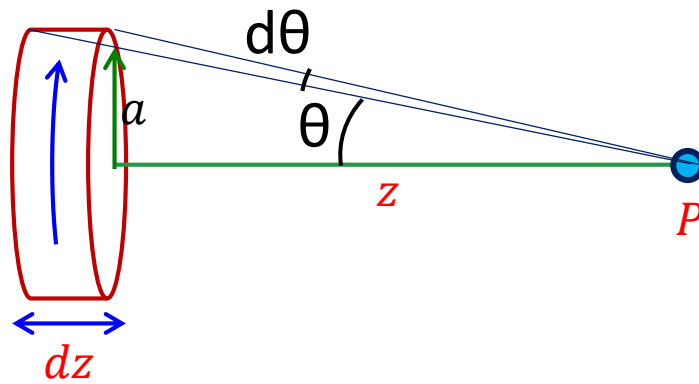
Magnetic field due to a circular ring of width  $dz$  , carries current  $I$  and radius  $a$  is ,

$$B = \frac{I \mu_o}{2} \int \frac{a^2}{(a^2 + z^2)^{3/2}} dz$$

For 'n' rings total current  $I' = nI$

Magnetic field due to solenoid

$$B = \frac{I' \mu_o}{2} \int \frac{a^2}{(a^2 + z^2)^{3/2}} dz$$



$$B = \frac{I' \mu_o}{2} \int \frac{a^2}{(a^2 + z^2)^{3/2}} dz$$

Using the geometry

$$\sin \theta = \frac{a}{(a^2 + z^2)^{1/2}}$$

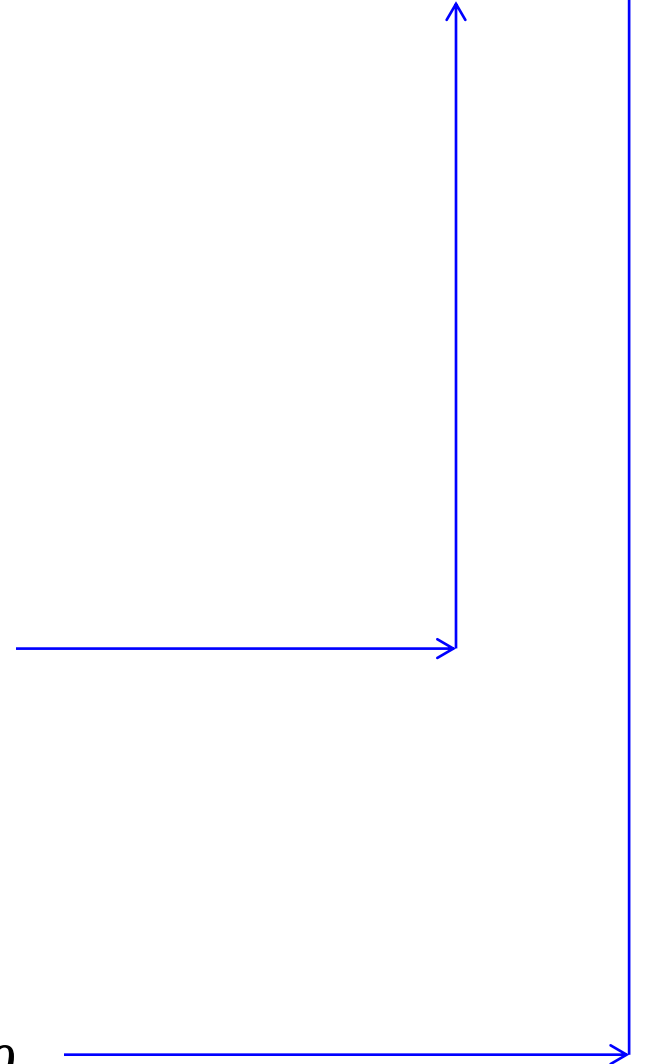
$$(a^2 + z^2)^{1/2} = \frac{a}{\sin \theta}$$

$$(a^2 + z^2)^{3/2} = \frac{a^3}{\sin^3 \theta}$$

And

$$z = \frac{a}{\tan \theta}$$

$$dz = -\frac{a}{\sin^2 \theta} d\theta$$

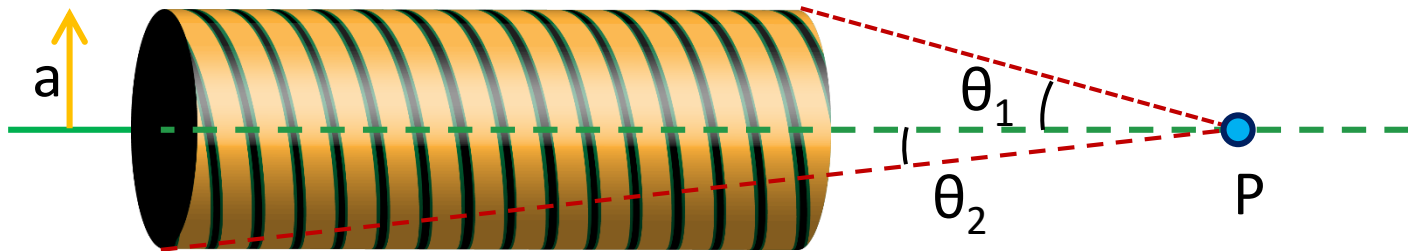


$$(a^2 + z^2)^{3/2} = \frac{a^3}{\sin^3 \theta}$$

$$dz = -\frac{a}{\sin^2 \theta} d\theta$$

Substituting these  
values  
In this equation

$$B = \frac{I' \mu_o}{2} \int \frac{a^2}{(a^2 + z^2)^{3/2}} dz$$



$$B = \frac{\mu_o n I}{2} \int_{\theta_1}^{\theta_2} \frac{a^2 \sin^3 \theta}{a^3 \sin^2 \theta} (-a d\theta)$$

$$B = \frac{\mu_o n I}{2} \int_{\theta_1}^{\theta_2} \frac{a^2 \sin^3 \theta}{a^3 \sin^2 \theta} (-a d\theta)$$

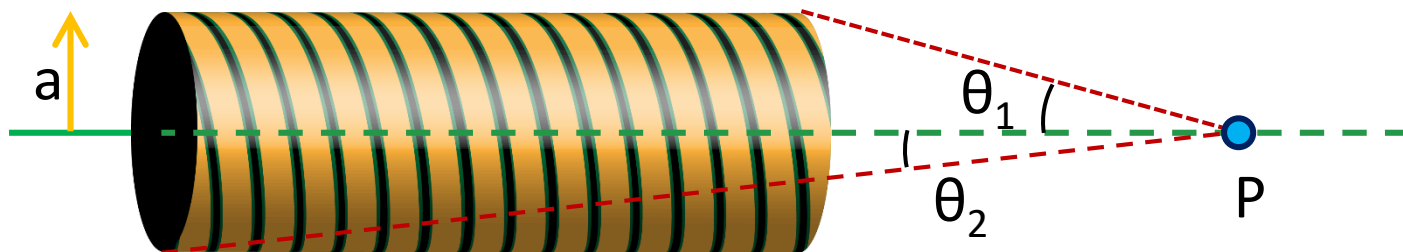
$$B = \frac{\mu_o n I}{2} (\cos \theta_2 - \cos \theta_1)$$

$$B = \frac{\mu_o n I}{2} (\cos\theta_2 - \cos\theta_1)$$

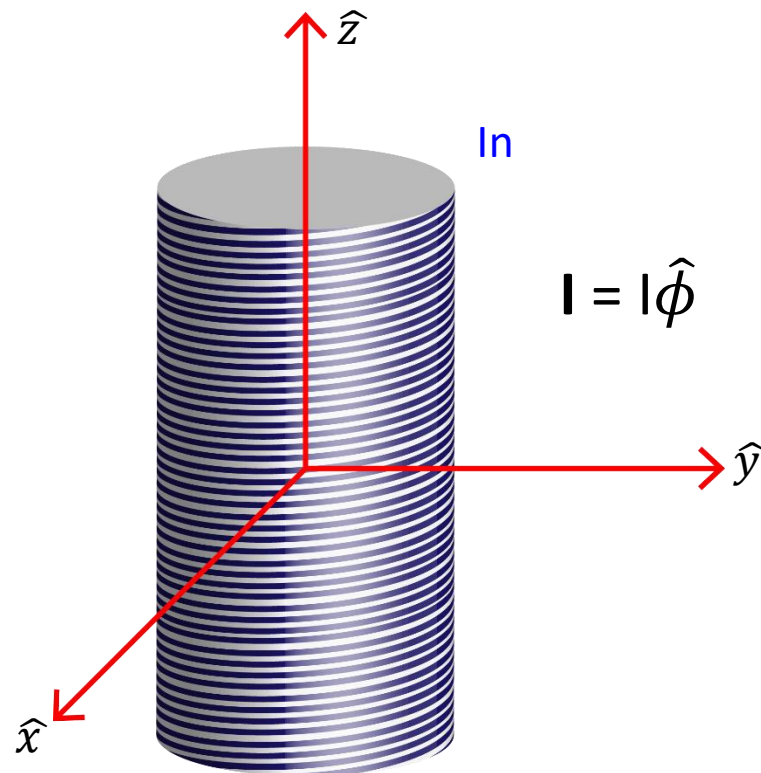
For an infinite solenoid  $\theta_2 = 0$  ,  $\theta_1 = \pi$

$$(\cos\theta_2 - \cos\theta_1) = 2$$

$$B = \mu_o n I$$



**Example-2:** Determine the magnetic field associated with an infinitely long solenoid consisting of  $n$  turns (loops) per unit length, i.e.  $n = N/L$  (where  $N$  = total number of turns or wire wound around a cylinder of radius  $R$  and carrying a steady current  $I$ ).



We will take several Amperian-loop contours in order to “explore” the nature of  $B$  in solenoid

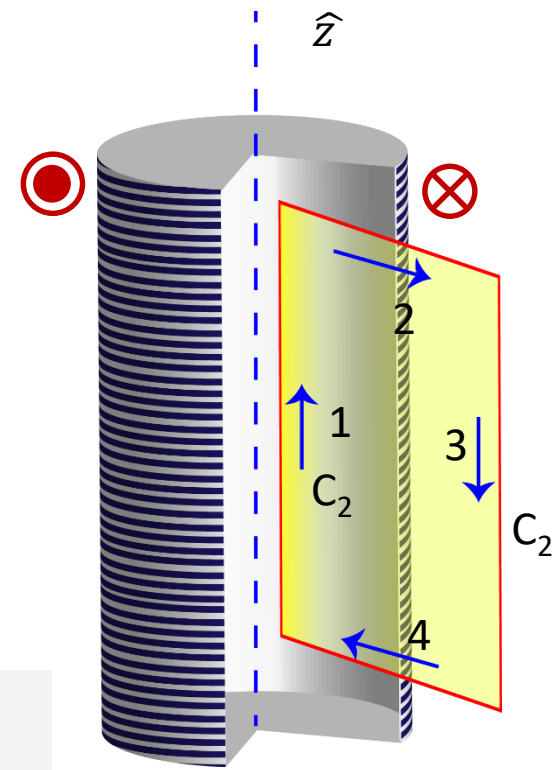
## Case-I : $r < R$ inside the solenoid

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$N_{\text{enclosed}} = nl$$

$$I_{\text{enclosed}} = N_{\text{enclosed}} I = n l I$$

Closed contour integral by breaking It up into Segments 1, 2,3 and 4.



$$\oint_{C_2} \vec{B}(r) \cdot d\vec{l} = \int_1 \vec{B}_1 \cdot d\vec{l}_1 + \int_2 \vec{B}_2 \cdot d\vec{l}_2 + \int_3 \vec{B}_3 \cdot d\vec{l}_3 + \int_4 \vec{B}_4 \cdot d\vec{l}_4$$

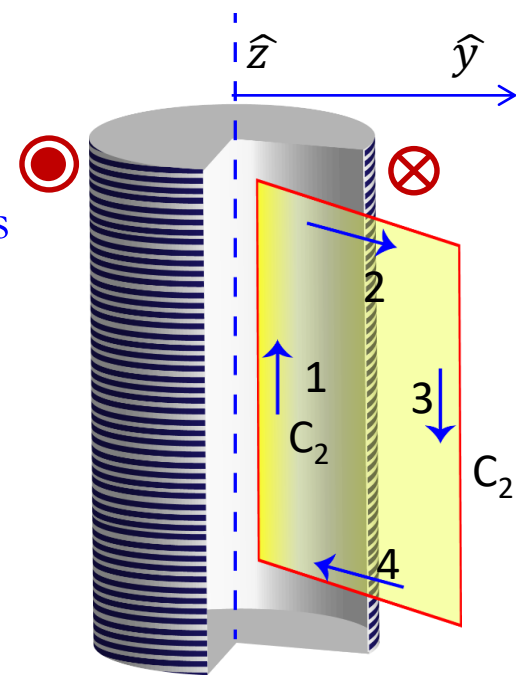
Here

$$dl_1 = +dz \hat{z}$$

$$dl_2 = +dy \hat{y} = +dr \hat{r} \quad \leftarrow \text{in cylindrical coordinates}$$

$$dl_3 = dz(-\hat{z}) = -dz \hat{z}$$

$$dl_4 = dy(-\hat{y}) = -dy \hat{y} = -dr \hat{r}$$



$$\oint_{C_2} \vec{B}(r) \cdot d\vec{l} = \int_{z=z_1}^{z_2} B_z(r < R) dz + \int_{r=R}^{r=\infty} B_2(r) \cdot dl_2$$

$$+ \underbrace{\int_{z=z_2}^{z=z_1} B_3(r = \infty) \cdot (dz \hat{z})}_{\text{Zero}} - \int_{r=\infty}^{r=R} \vec{B}_4(r) \cdot dr \hat{r}$$

at  $r \rightarrow \infty, B = 0$

$$N_{\text{enclosed}} = nl$$

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{\text{enclosed}}$$

$$I_{\text{enclosed}} = N_{\text{enclosed}} I = n l I$$

$$B_z(r \leq R)l = \mu_0 n l I \quad \Rightarrow \quad \vec{B}_z(r \leq R) = \mu_0 n I$$

Thus we learn that the magnetic field is only non-zero *inside* ( $r < R$ ) the infinitely long solenoid and that it points in the  $\hat{z}$  - direction:

$$\vec{B}_z(r \leq R) = \mu_0 n I$$

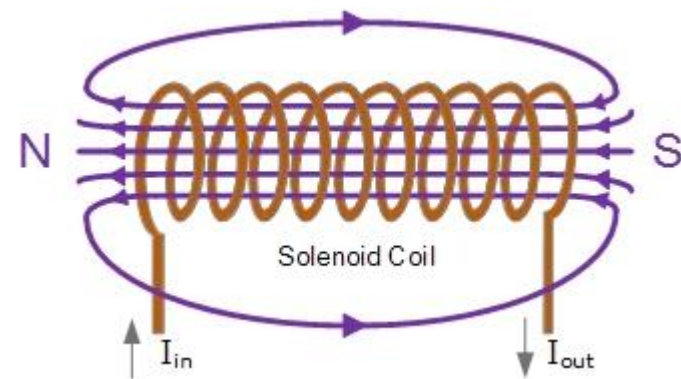
We can also calculate the magnetic flux passing through the bore of the solenoid:

$$\Phi_m = \int \vec{B}_z(r \leq R) \cdot d\vec{A}$$

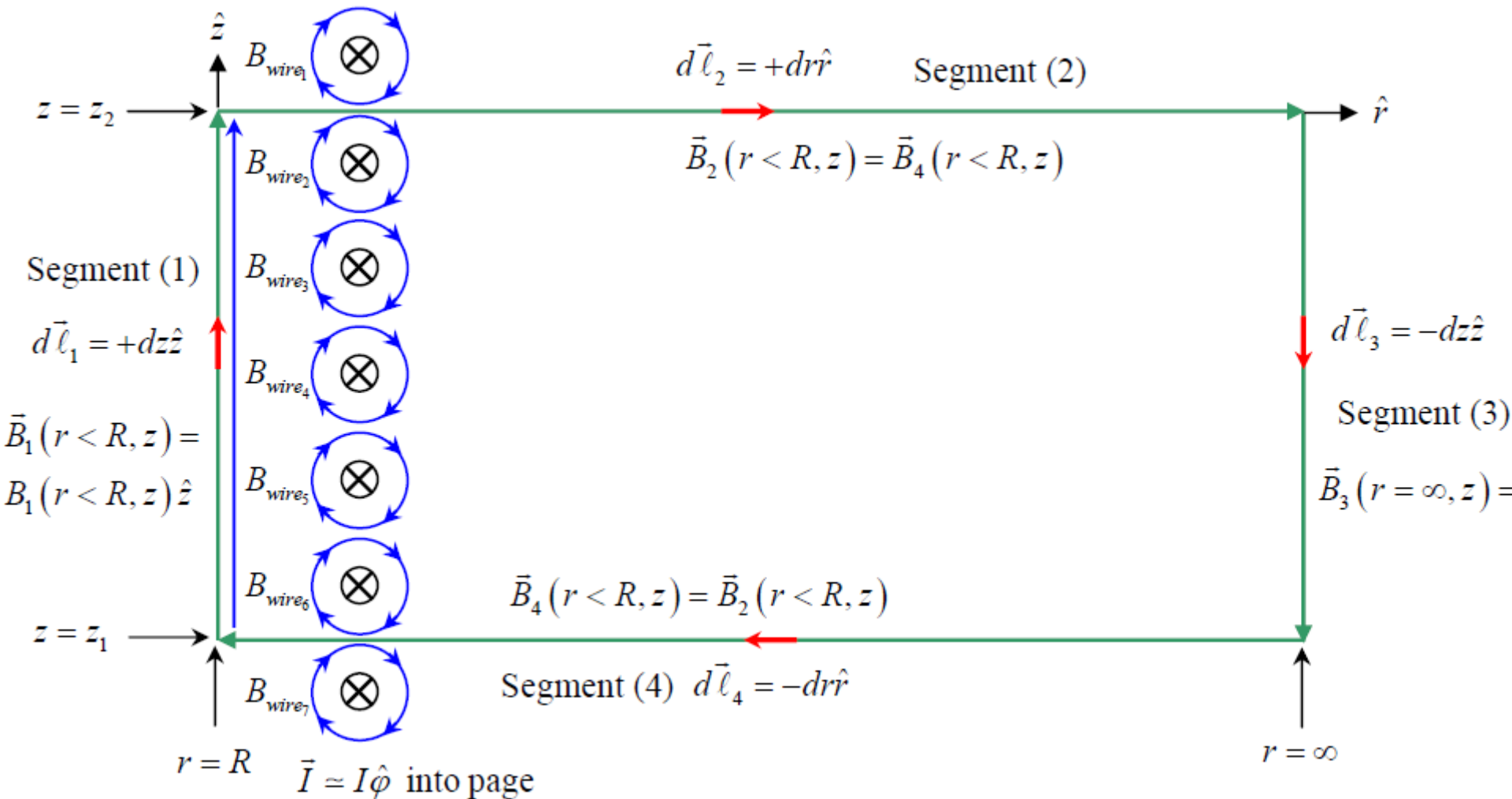
$$d\vec{A} = dA \hat{n} = dA \hat{z}$$

$$\Phi_m = \int \vec{B}_z(r \leq R) \cdot d\vec{A}$$

$$\Phi_m = \int \vec{B}_z(r \leq R) \cdot dA \hat{z} = \mu_0 n I \pi R^2$$



We will extend the RHS of contour  $C_2$  out to infinity – i.e. in the radial direction (in cylindrical coordinates) such that segment (3) of contour  $C_2$  is located at  $r = \infty$ , as shown in a close-up view of the solenoid & contour integral's path, below:



Case-II :  $r > R$  outside the solenoid

$C_1$  Amperian loop in XY plane

$$\oint_{C_1} \vec{B}_{outside} \cdot d\vec{l} = \mu_0 I_{enclosed}$$

$$d\vec{l} = r d\phi \hat{\phi}$$

$$I = I \hat{\phi}$$

$$\int_0^{2\pi} \vec{B}_{outside} \cdot (r d\phi \hat{\phi}) = \mu_0 I_{enclosed}$$

$$B_{outside} 2\pi r = \mu_0 I$$

Encloses a single winding of the solenoid

$$\vec{B}_{outside} = \frac{\mu_0 I}{2\pi r} \hat{\phi}$$



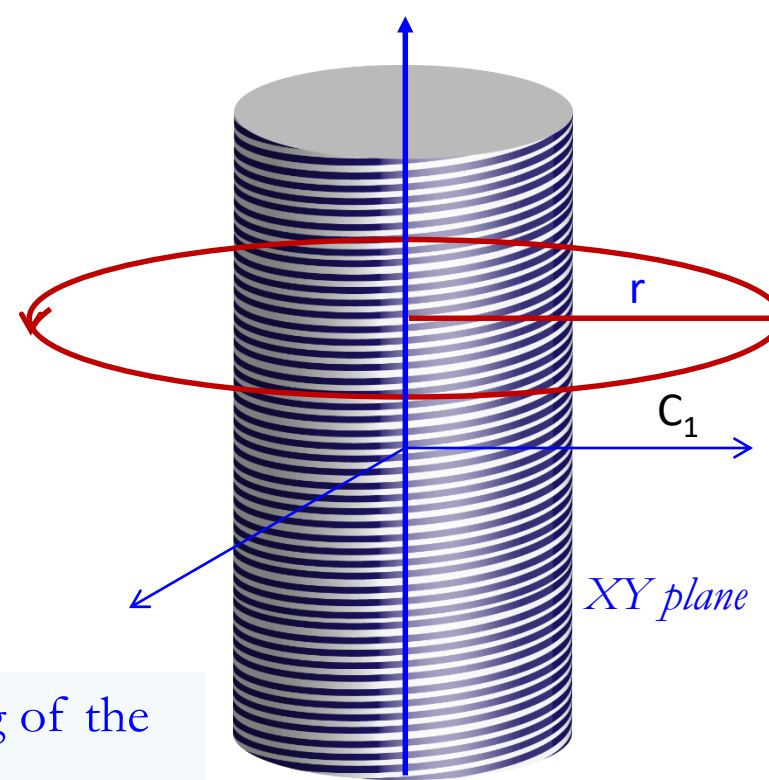
Similar to infinite long wire

It arises solely due to finite pitch angle ( $\alpha$ ) associated with physically winding a real solenoid.

If we neglect the finite pitch angle of the solenoid windings, i.e.  $\alpha = 0$ , then



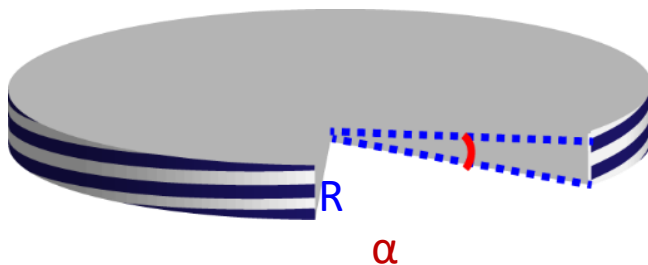
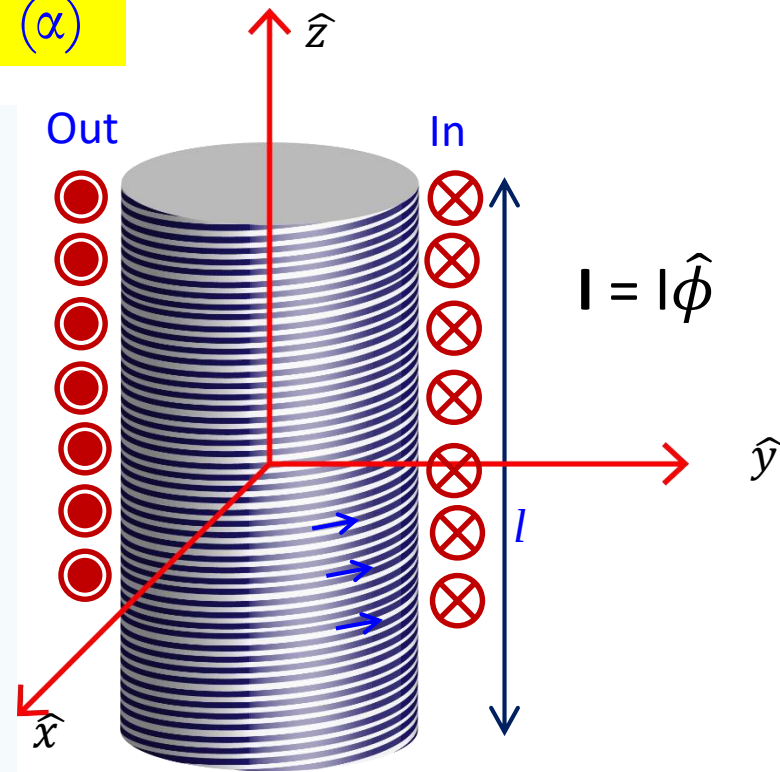
$$\vec{B}_{outside} = 0$$



## Finite pitch angle ( $\alpha$ )

There is a very small pitch angle,  $\alpha$  associated with physically winding real, finite diameter wire around a cylindrical form to make a real solenoid coil.

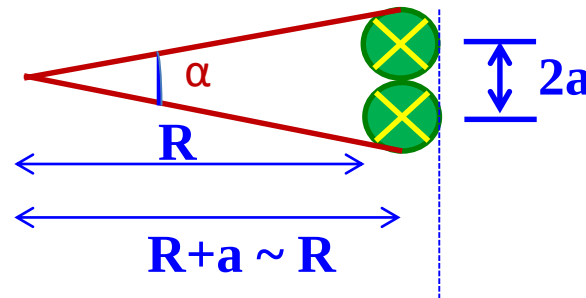
The steady current flowing in the solenoid is then  $\mathbf{I} = I \cos\alpha \hat{\phi} + I \sin\alpha \hat{z}$ , which for  $\alpha \approx 0$  is nearly all in the phi direction, i.e.  $\mathbf{I} = I \hat{\phi}$ .



If we neglect this finite-small pitch angle  $\alpha$ , then  $\mathbf{I} \equiv I \hat{\phi}$ .

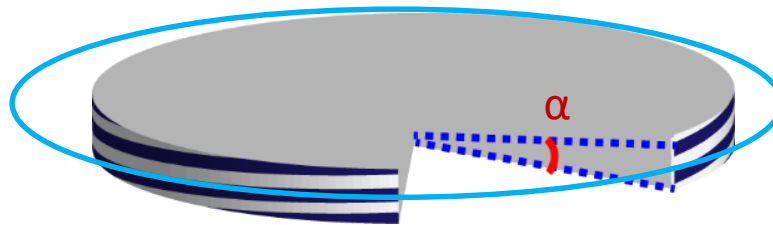
# Close-up/Cutaway View

Finite pitch angle,  $\alpha$



$$I_{\text{outside}} = 0$$

$$\vec{B}_{\text{outside}} = 0$$



The finite pitch angle  $\alpha$  associated with the windings of a solenoid coil arises due to winding the coil with wire of finite thickness (e.g. wire radius  $= a$ ).

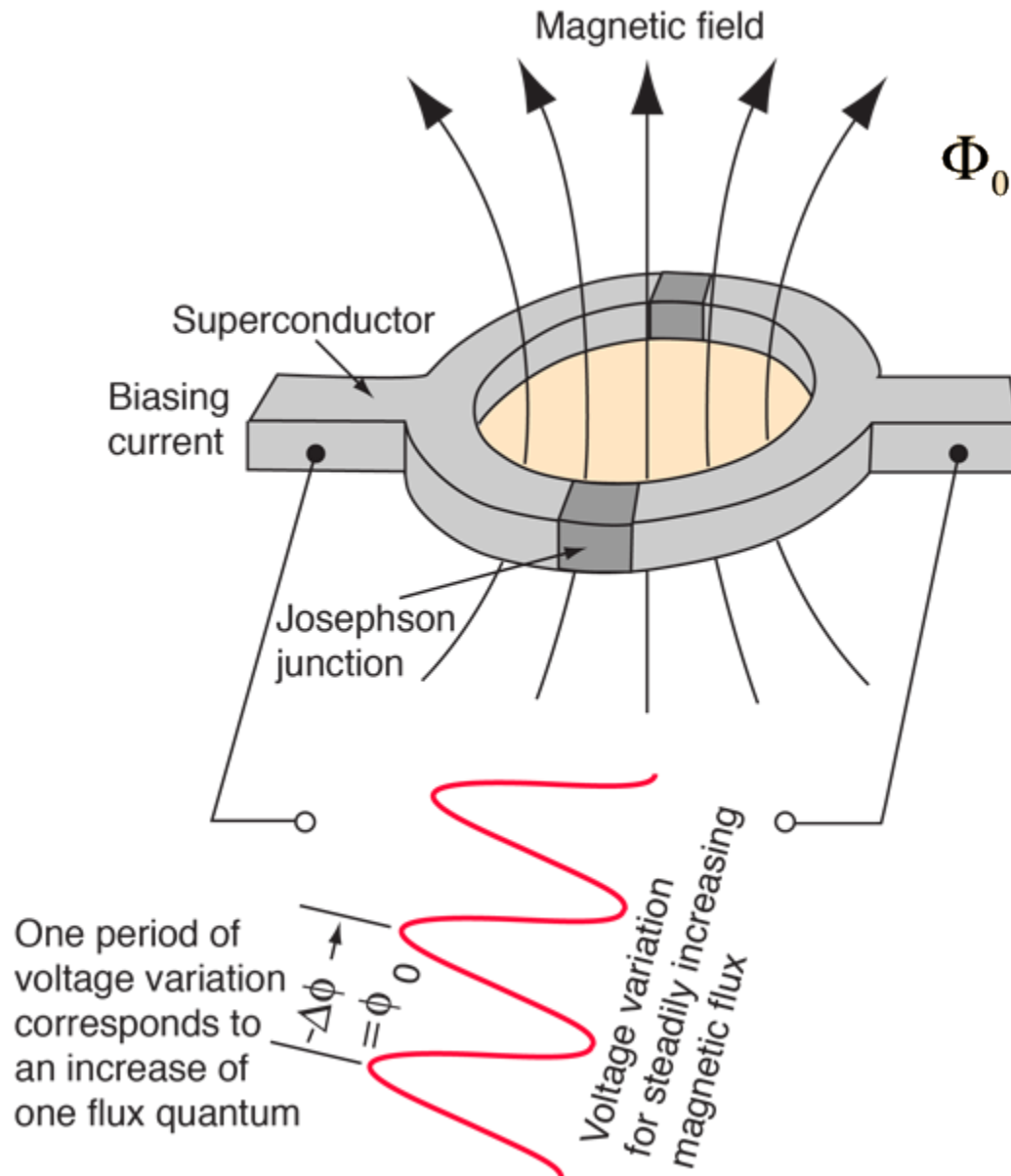
## Digital Gaussmeter measuring Magnetic field intensity 'H' (A/m)



Operates on the principle of Hall Effect in Semiconductors. A semiconductor carrying current develops an electromotive force, when placed in a magnetic field, in a direction perpendicular to the direction of both electric current and magnetic field. The magnitude of this e.m.f. is proportional to the field intensity, if the current is kept constant. This e.m.f. is called the Hall voltage. The small Hall Voltage is amplified through a high stability amplifier so that a Millivoltmeter connected at the output of the amplifier can be calibrated directly in magnetic field unit (gauss).

# Applications

# SQUID Magnetometer



$$\Phi_0 = \frac{2\pi\hbar}{2e} \cong 2.0678 \times 10^{-15} \text{ tesla} \cdot \text{m}^2$$

If a constant biasing current is maintained in the SQUID device, the measured voltage oscillates with the changes in phase at the two junctions, which depends upon the change in the magnetic flux. Counting the oscillations allows you to evaluate the flux change which has occurred.

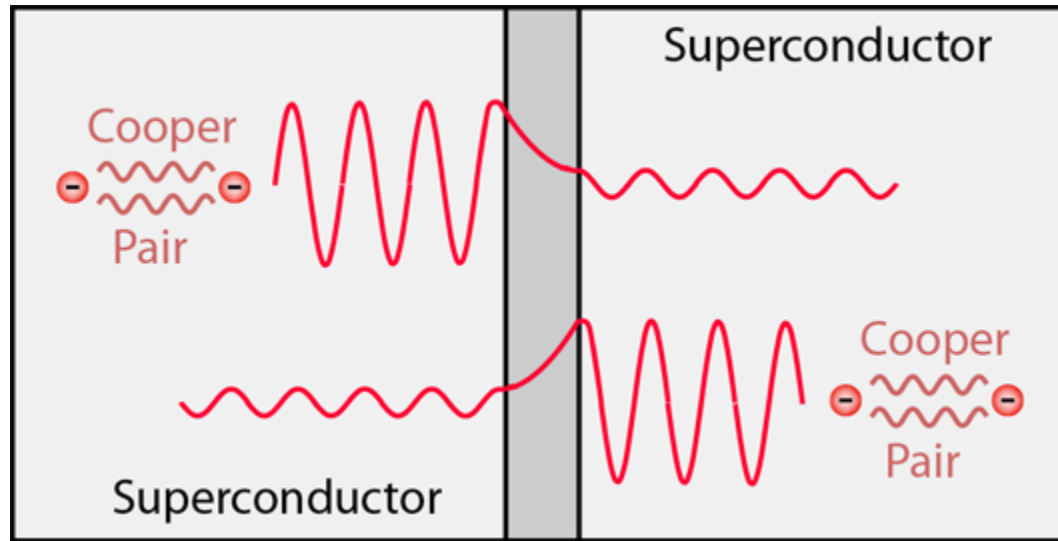
# SQUID Magnetometer

The superconducting quantum interference device (SQUID) consists of two superconductors separated by thin insulating layers to form two parallel [Josephson junctions](#). The device may be configured as a magnetometer to detect incredibly small [magnetic fields](#) -- small enough to measure the magnetic fields in living organisms. Squids have been used to measure the magnetic fields in mouse brains to test whether there might be enough magnetism to attribute their navigational ability to an internal compass

The great sensitivity of the SQUID devices is associated with measuring changes in magnetic field associated with one flux quantum. One of the discoveries associated with Josephson junctions was that flux is quantized in units

|                          |              |
|--------------------------|--------------|
| Threshold for SQUID:     | $10^{-14}$ T |
| Magnetic field of heart: | $10^{-10}$ T |
| Magnetic field of brain: | $10^{-13}$ T |

# Josephson Junction



The wavefunction which describes a Cooper pair of electrons in a superconductor is an exponential like the free particle wavefunction. In fact, all the Cooper pairs in a superconductor can be described by a single wavefunction in the absence of a current because all the pairs have the same phase - they are said to be "phase coherent" (Clarke). If two superconductors are separated by a thin insulating layer, then quantum mechanical [tunneling](#) can occur for the Cooper pairs without breaking up the pairs. Clarke envisions this condition as the wavefunctions for Cooper pairs on each side of the junction penetrating into the insulating region and "locking together" in phase. Under these conditions, a current will flow through the junction in the absence of an applied voltage (the DC Josephson effect).